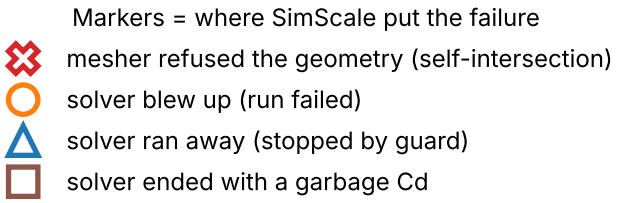
Kirigami parachute: every geometry tried and where it failed

43 runs on 17 distinct shapes; 9 finished with a valid Cd. Canopy builds: legacy (original), clean (fused slabs), lattice (pinned joints).



Outcomes

ok: 9 (0002, 0005, 0009, 0011, 0013, 0023, 0025, 0028, 0043)

solver diverged (stopped by guard): 9 (0021, 0024, 0030, 0033, 0034, 0036, 0037, 0038, 0039)

mesher failed: 6 (0003, 0004, 0006, 0022, 0026, 0027)

solver diverged (crashed): 6 (0007, 0012, 0017, 0031, 0032, 0035)

solver diverged (garbage Cd): 6 (0008, 0014, 0015, 0016, 0018, 0020)

not run (cost estimate over the cap): 1 (0001)

pipeline / API error (not the geometry): 1 (0010)

solver not converged: 1 (0019)

mesh cap hit (cancelled): 1 (0029)

other: SimScaleError: mesh quality gate: 583 tetrahedra with non-or: 1 (0040)

other: SimScaleError: mesh quality gate: 376 tetrahedra with non-or: 1 (0041)

other: SimScaleError: simulation run cancelled: used 4.53 CPU h, ov: 1 (0042)

What the positions say

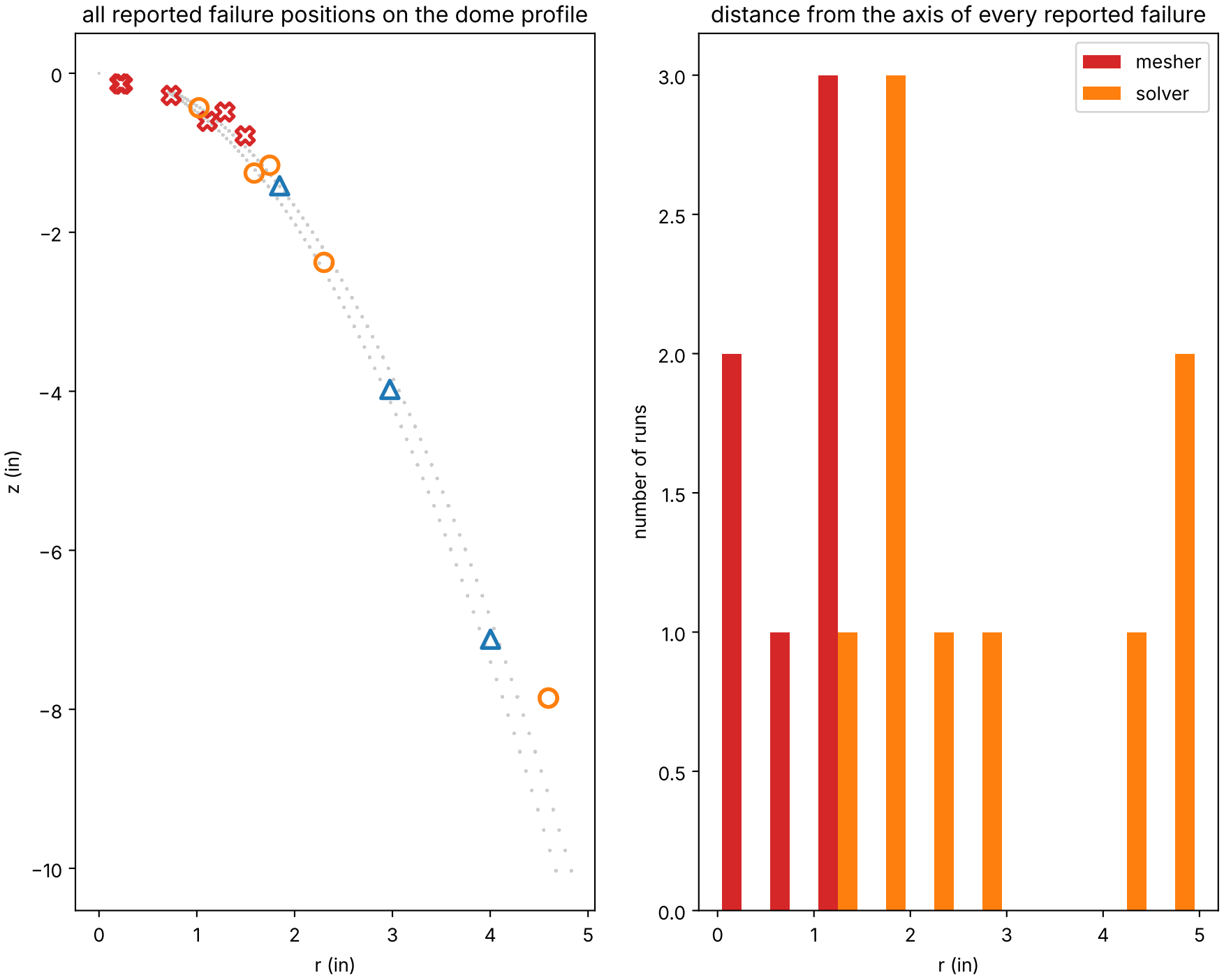
- Mesher failures (red X) all sit within 1.5 in of the apex: the centre cap and the first rings (the old builds' crossing triangles and slivers).

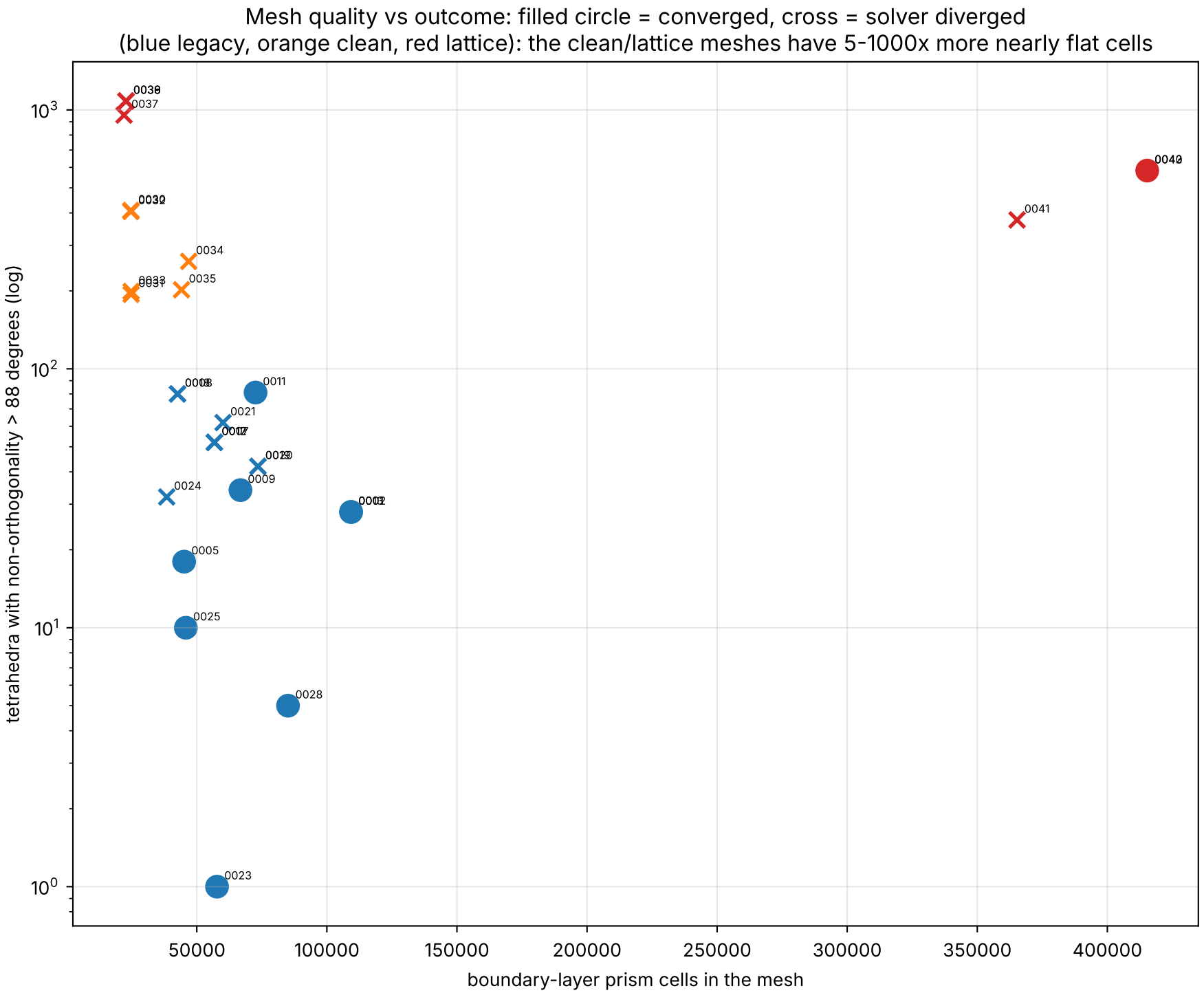
- Solver blow-ups (orange / blue) are spread over r 1.0 - 4.6 in, on a different ribbon in each run; they follow the same designs across builds, not one spot.

- Runs 0001, 0010, 0014-0016 and 0019 were pipeline / settings problems, not geometry; 0029 hit the mesh CPU cap.

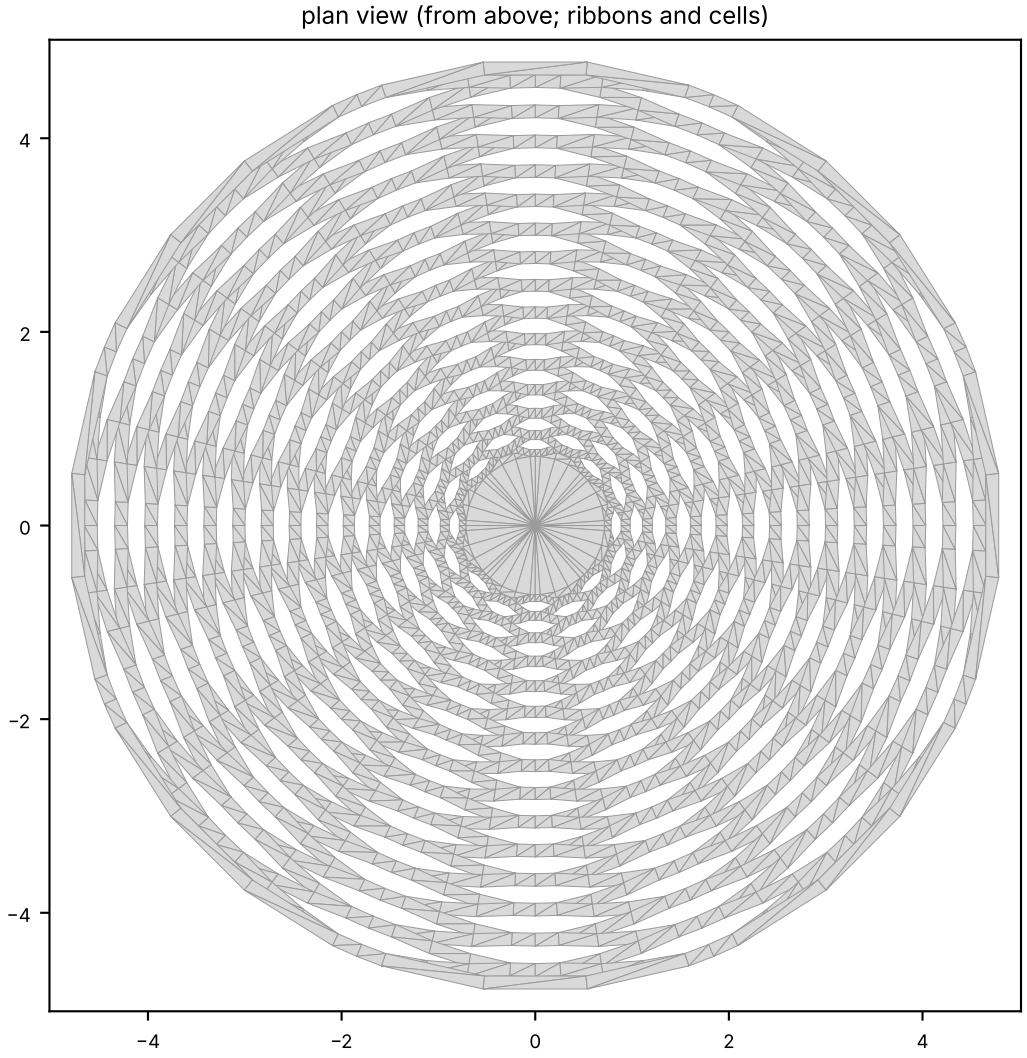
- Valid Cds (8 runs) all came from the legacy build; no clean- or lattice-build run has converged yet.

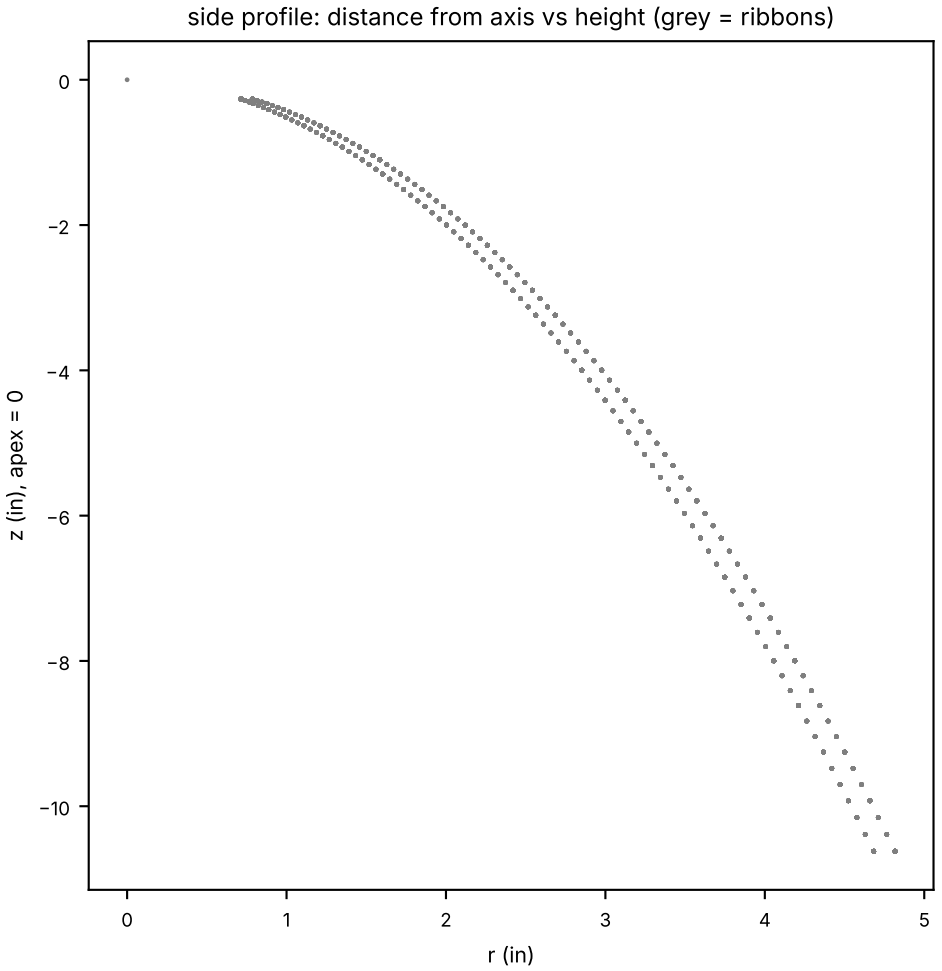
- Mesh quality is the common factor: converged runs had at most 81 nearly-flat tetrahedra (non-orthogonality > 88 deg); clean/lattice meshes had 190-1100 and half the boundary-layer cel


all reported failure positions on the dome profile
z (in)
r (in)
0
−2
−4
−6
−8
−10
0
1
2
3
4
5
distance from the axis of every reported failure
number of runs
r (in)
mesher
solver
0.0
0.5
1.0
1.5
2.0
2.5
3.0


Mesh quality vs outcome: filled circle = converged, cross = solver diverged
(blue legacy, orange clean, red lattice): the clean/lattice meshes have 5-1000x more nearly flat cells
tetrahedra with non-orthogonality > 88 degrees (log)
boundary-layer prism cells in the mesh
10^3
10^2
10^1
10^0
50000
100000
150000
200000
250000
300000
350000
400000
0037
0034
0035
0041
0011
0021
0009
0024
0005
0025
0028
0023

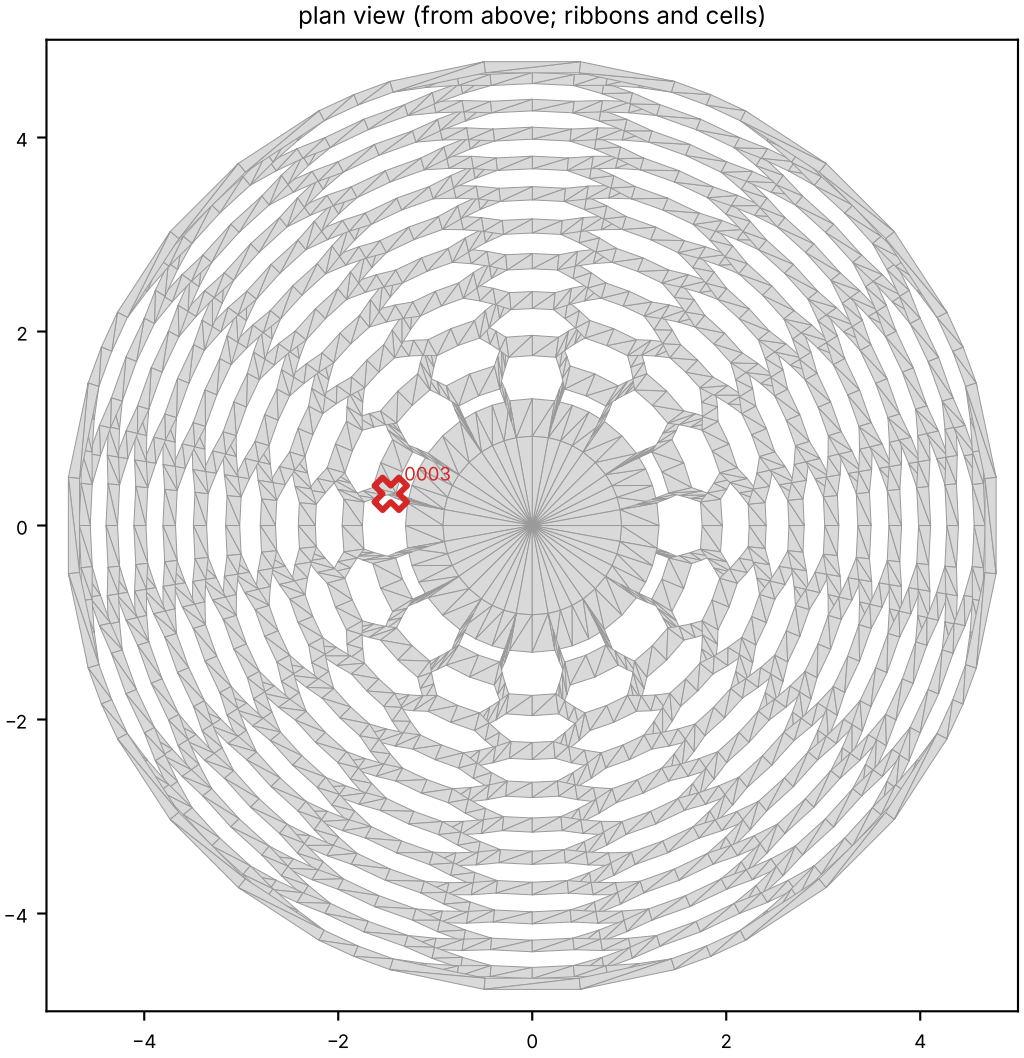
Shape 1: 30 rings, 8 cuts per ring, ring exponent 1.15, inner radius 0.75 in

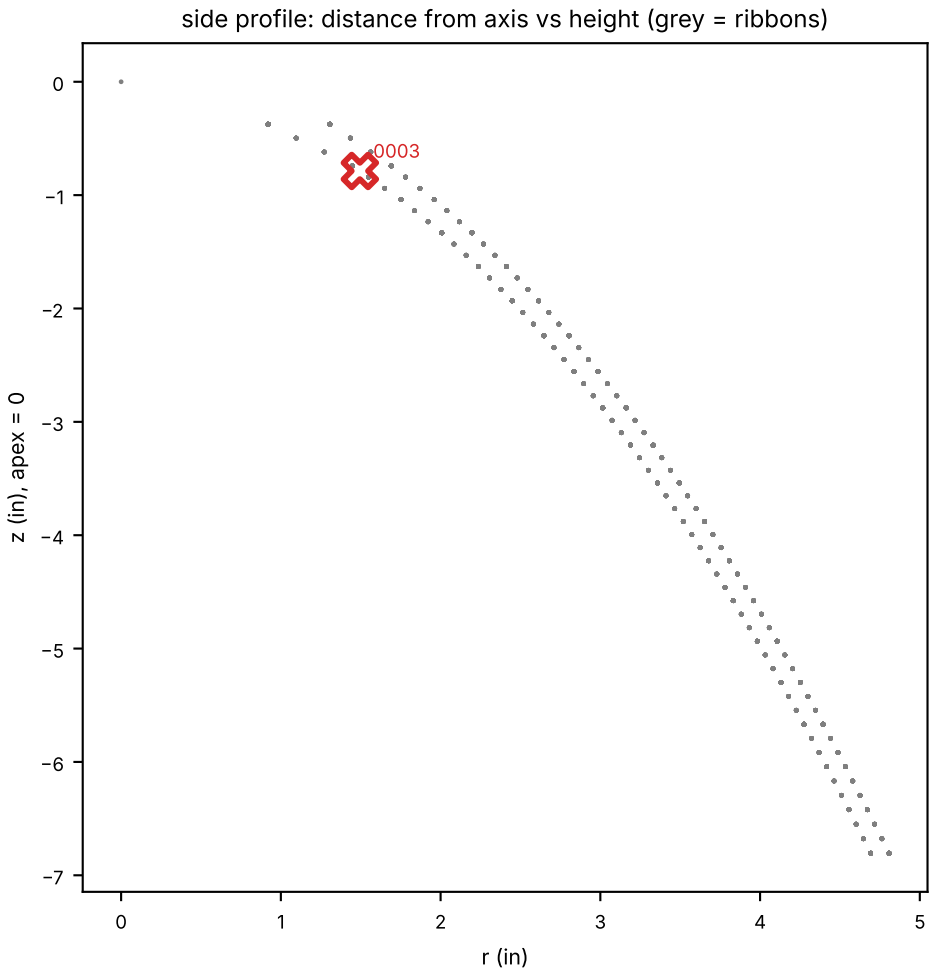




run	build	solver settings	outcome	CPU h	where / result
design_0001	legacy	fineness 4.5; 500 it; default nume	not run (cost estimate over the cap)	0.26	no position reported
design_0002	legacy	fineness 4.5; 500 it; default nume	ok	3.51	Cd 0.7327
design_0013	legacy	fineness 4.5; 500 it; potential-fl	ok	3.45	Cd 0.7537
design_0016	legacy	never ran: SimScale rejected the s	solver diverged (garbage Cd)	0.00	no position reported

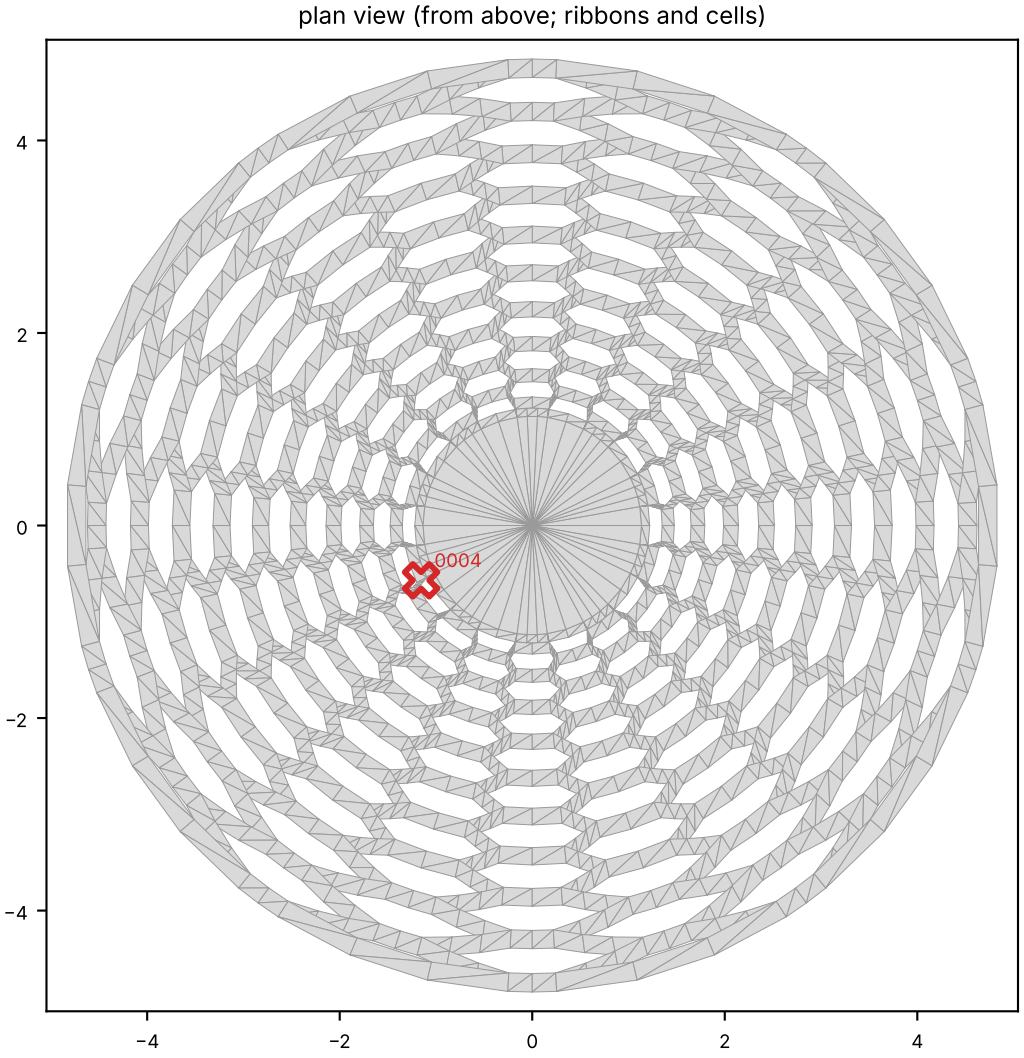
Shape 2: 20 rings, 8 cuts per ring, ring exponent 0.71, inner radius 1.11 in

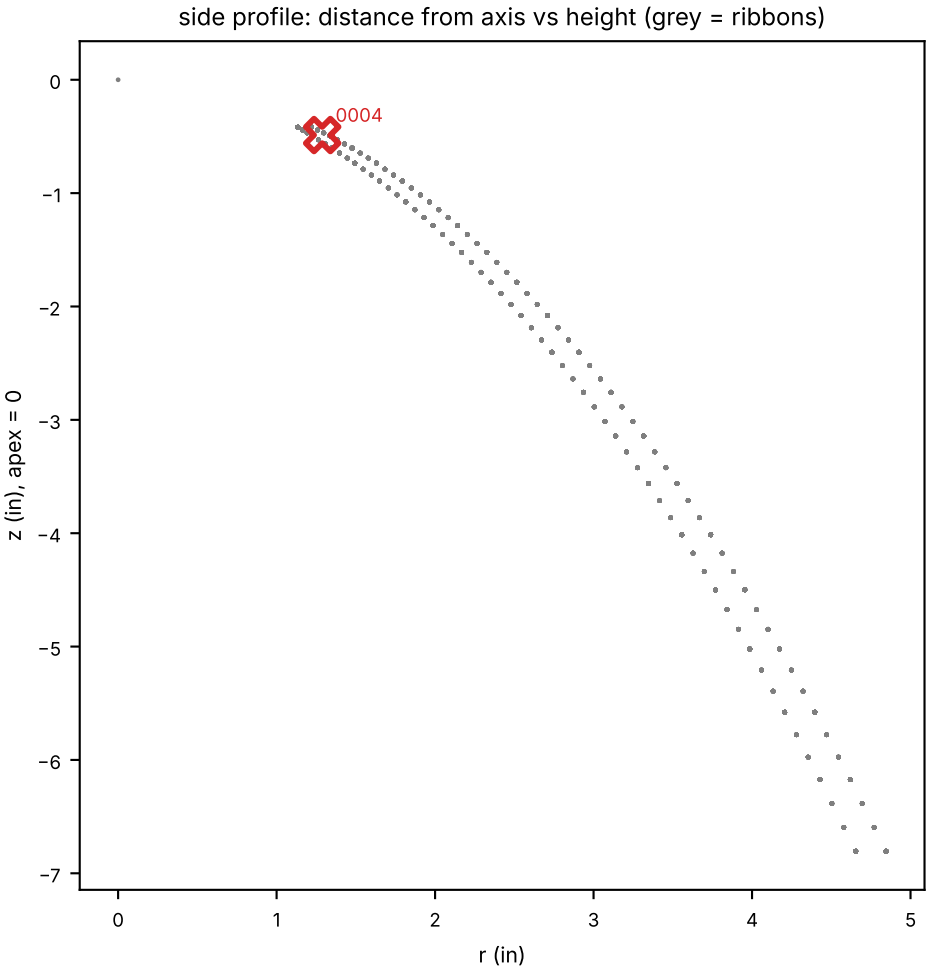




run	build	solver settings	outcome	CPU h	where / result
design_0003	legacy	fineness 4.5; 500 it; default nume	mesher failed	0.00	r 1.49 in, z -0.79 in

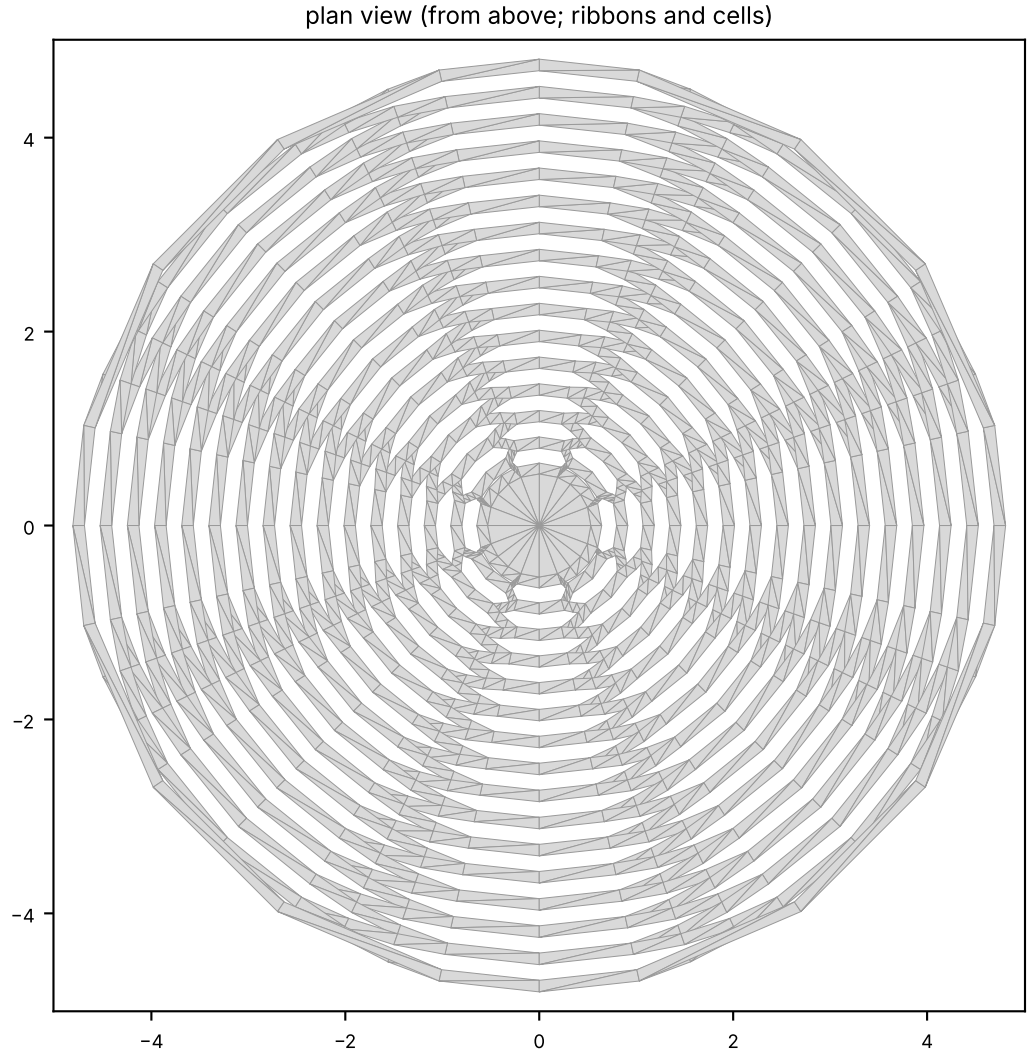
Shape 3: 20 rings, 10 cuts per ring, ring exponent 1.21, inner radius 1.18 in

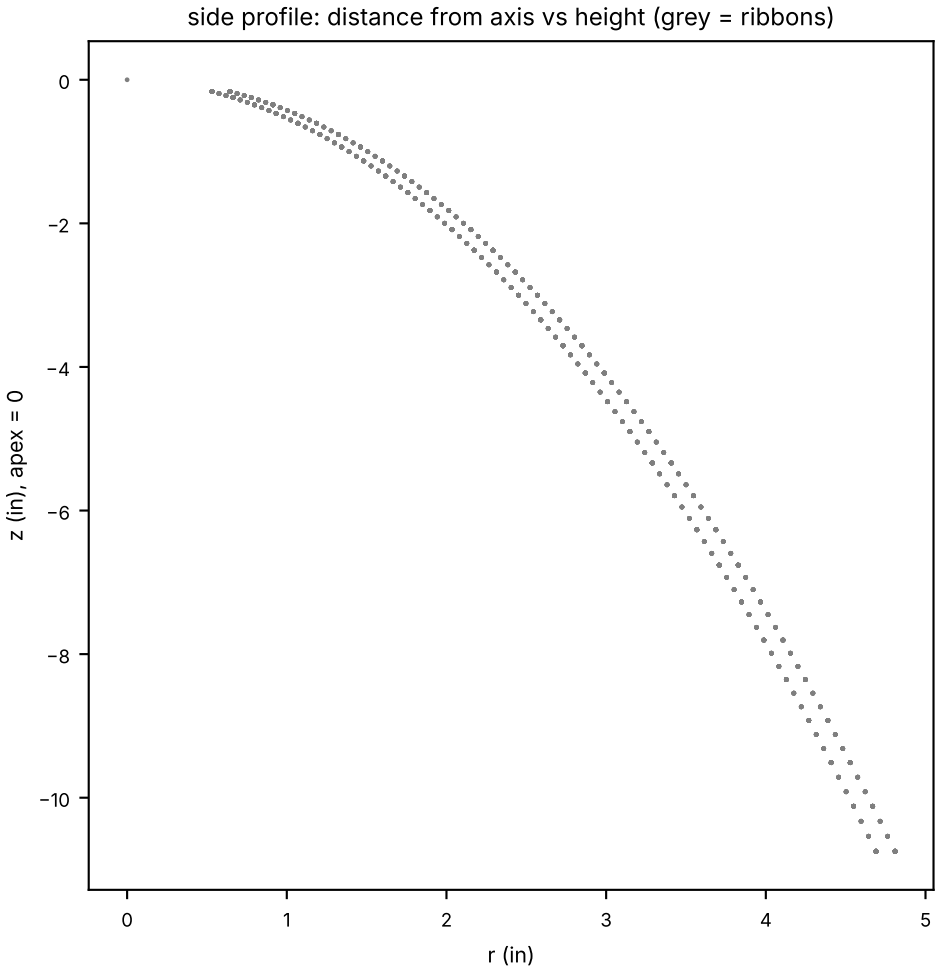




run	build	solver settings	outcome	CPU h	where / result
design_0004	legacy	fineness 4.5; 500 it; default nume	mesher failed	0.00	r 1.29 in, z -0.49 in

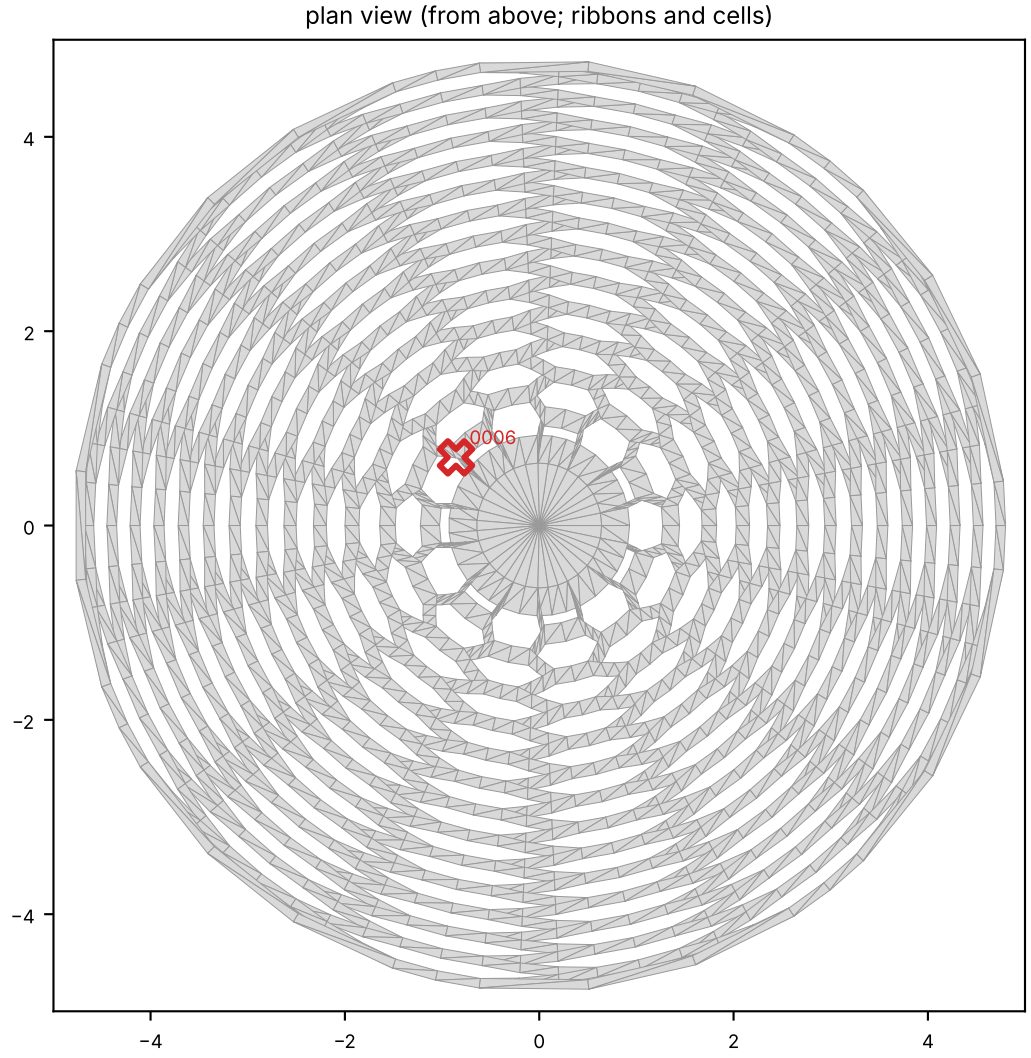
Shape 4: 31 rings, 4 cuts per ring, ring exponent 1.01, inner radius 0.59 in

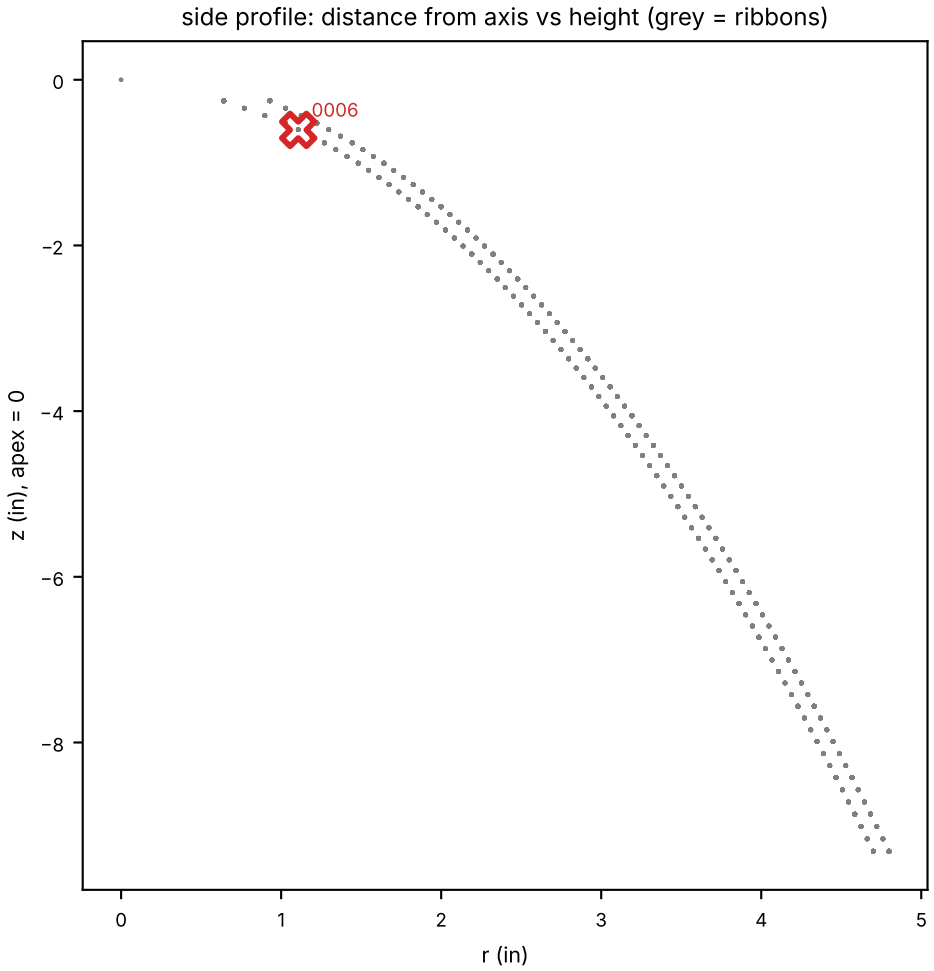




run	build	solver settings	outcome	CPU h	where / result
design_0005	legacy	fineness 4.5; 500 it; default nume	ok	2.90	Cd 0.7812

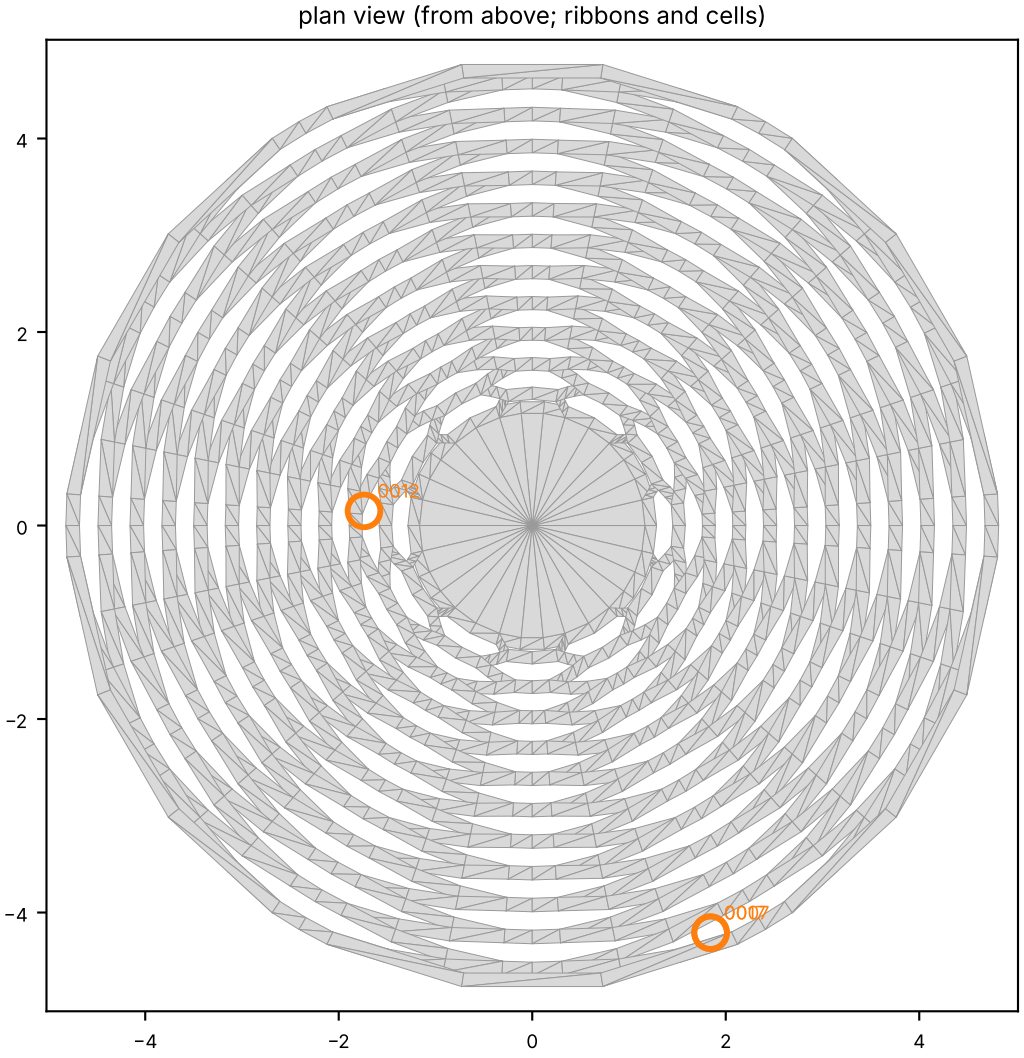
Shape 5: 27 rings, 7 cuts per ring, ring exponent 0.76, inner radius 0.79 in

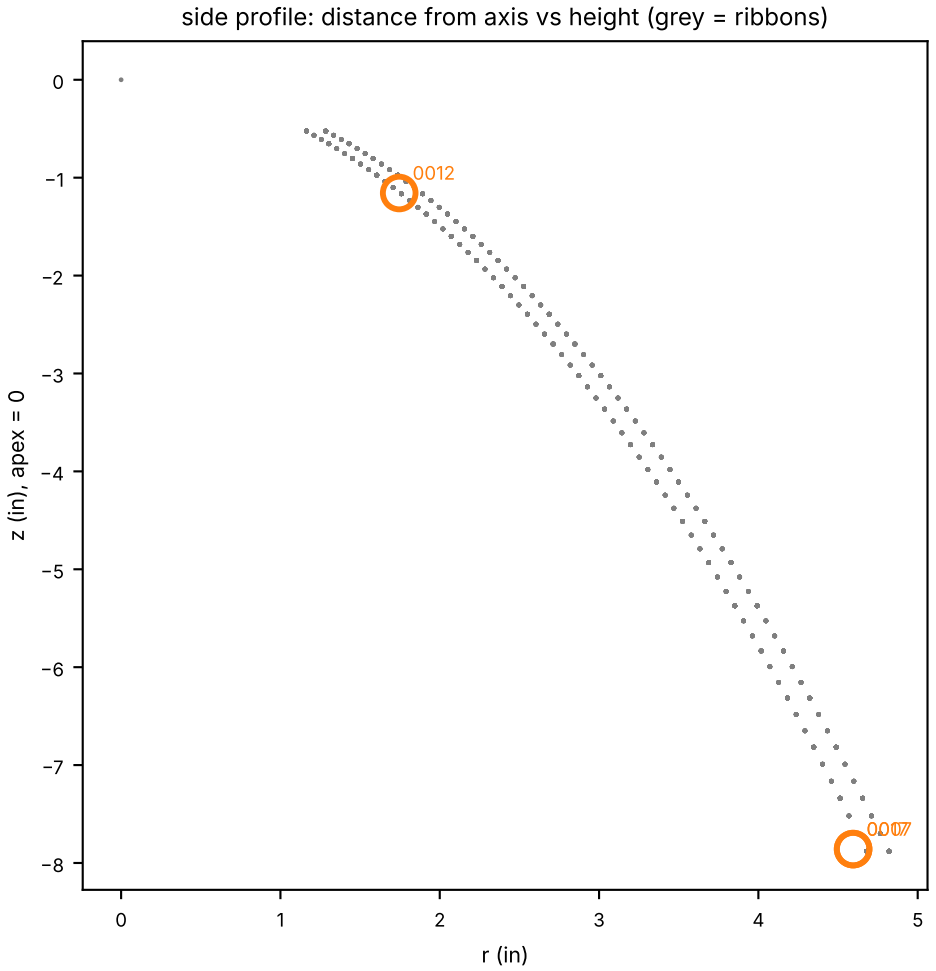




run	build	solver settings	outcome	CPU h	where / result
design_0006	legacy	fineness 4.5; 500 it; default nume	mesher failed	0.00	r 1.11 in, z -0.61 in

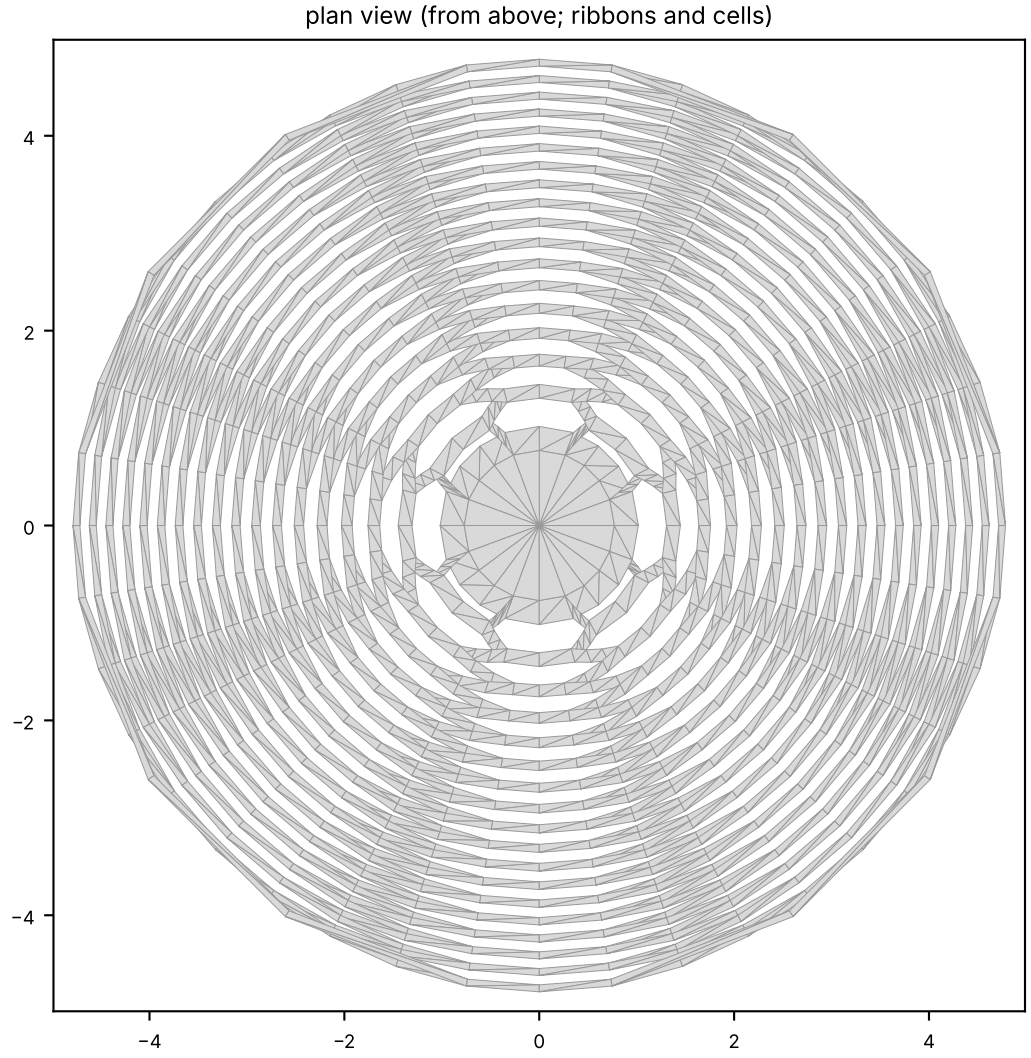
Shape 6: 23 rings, 6 cuts per ring, ring exponent 1.04, inner radius 1.22 in

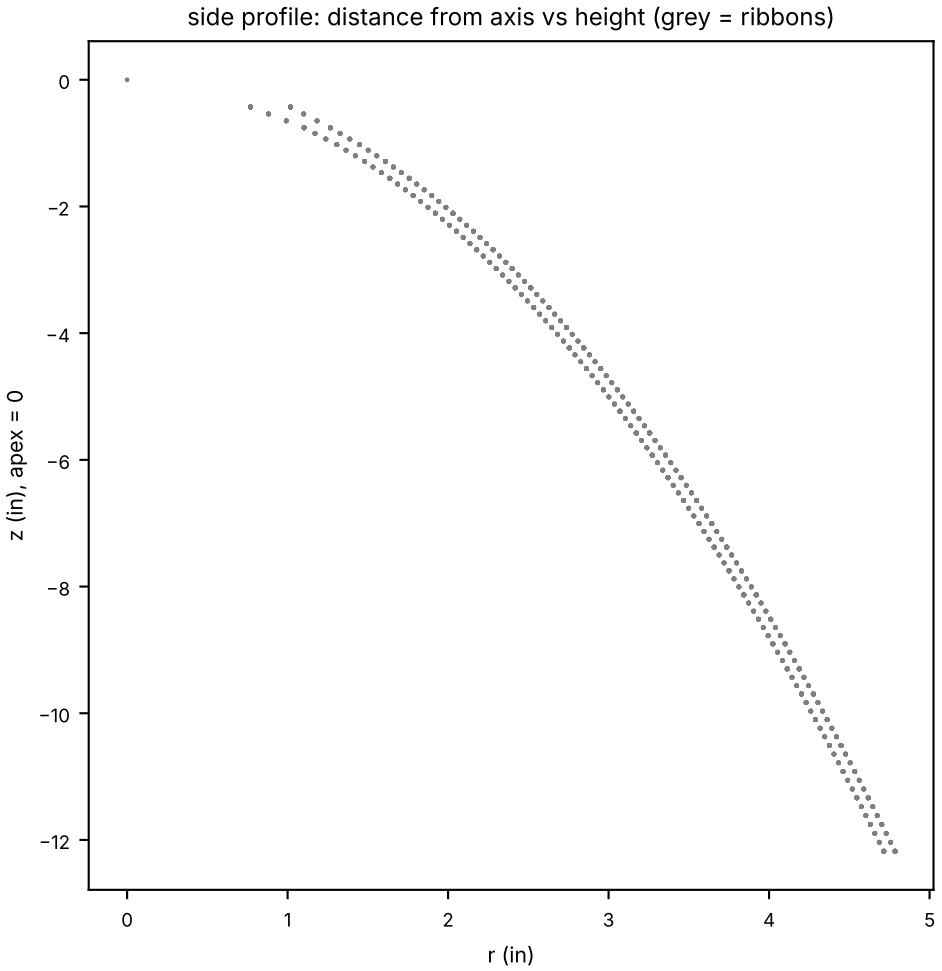




run	build	solver settings	outcome	CPU h	where / result
design_0007	legacy	fineness 4.5; 500 it; default nume	solver diverged (crashed)	2.58	r 4.59 in, z -7.86 in
design_0012	legacy	fineness 4.5; 500 it; potential-fl	solver diverged (crashed)	2.02	r 1.75 in, z -1.16 in
design_0014	legacy	never ran: SimScale rejected the s	solver diverged (garbage Cd)	0.00	no position reported
design_0017	legacy	fineness 4.5; 500 it; upwind k/ome	solver diverged (crashed)	2.31	r 4.59 in, z -7.86 in
design_0019	legacy	fineness 6; 500 it; default numeri	solver not converged	4.60	no position reported
design_0020	legacy	fineness 6; 1400 it; default numer	solver diverged (garbage Cd)	11.56	no position reported

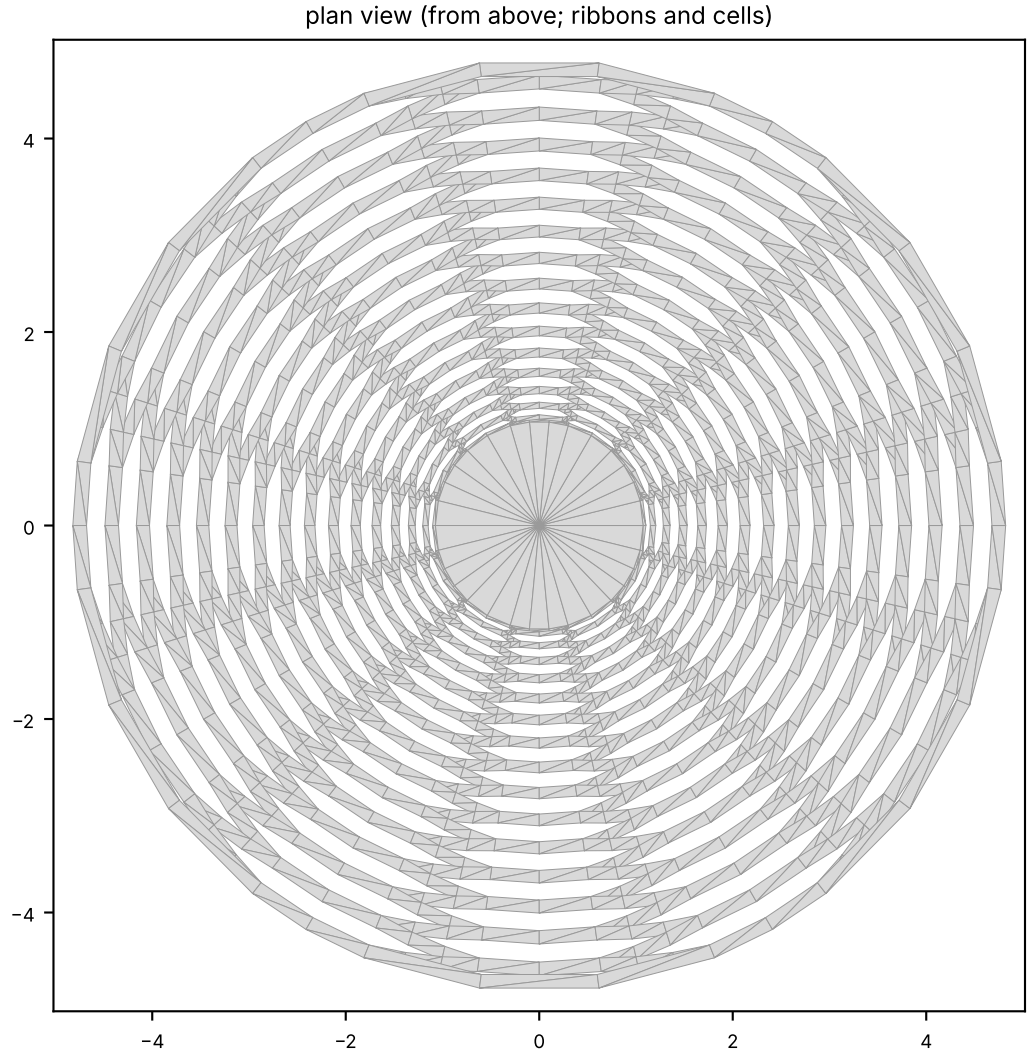
Shape 7: 35 rings, 4 cuts per ring, ring exponent 0.73, inner radius 0.89 in

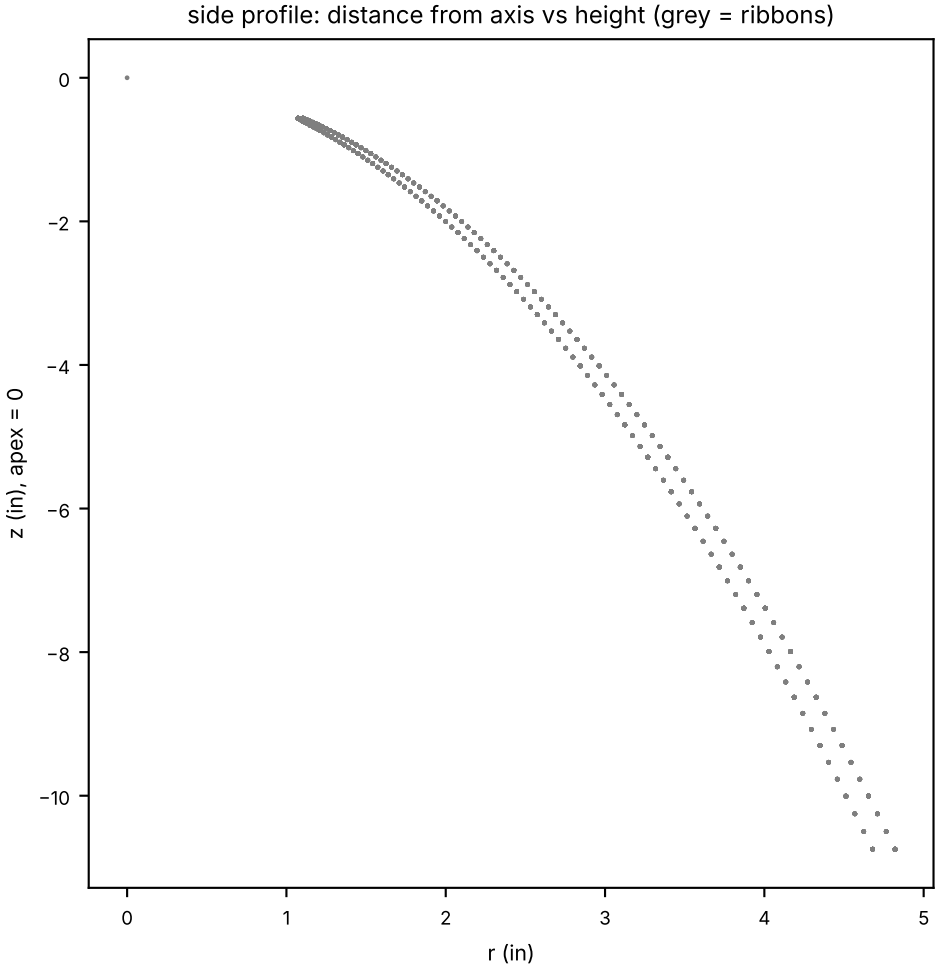




run	build	solver settings	outcome	CPU h	where / result
design_0008	legacy	fineness 4.5; 500 it; default nume	solver diverged (garbage Cd)	3.17	no position reported
design_0015	legacy	never ran: SimScale rejected the s	solver diverged (garbage Cd)	0.00	no position reported
design_0018	legacy	fineness 4.5; 500 it; upwind k/ome	solver diverged (garbage Cd)	0.56	no position reported

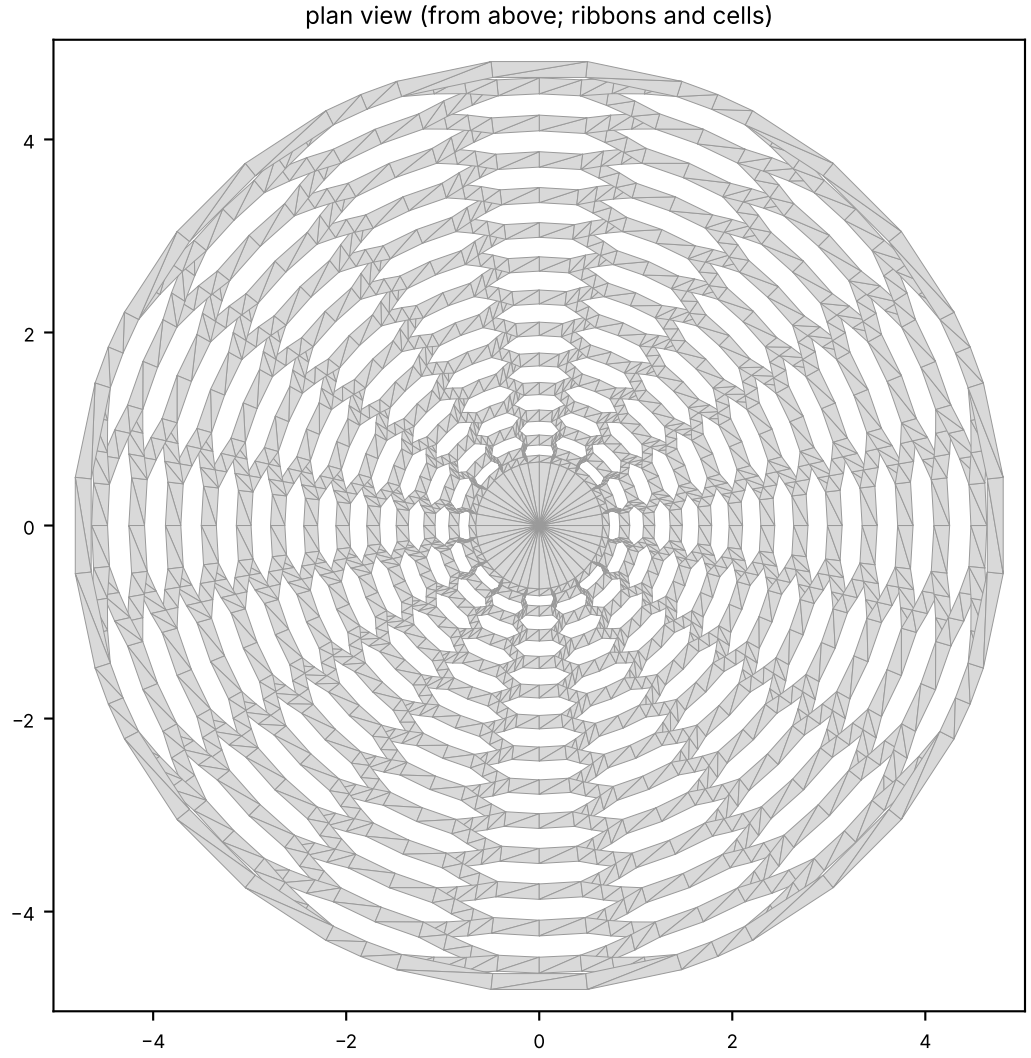
Shape 8: 31 rings, 6 cuts per ring, ring exponent 1.37, inner radius 1.09 in

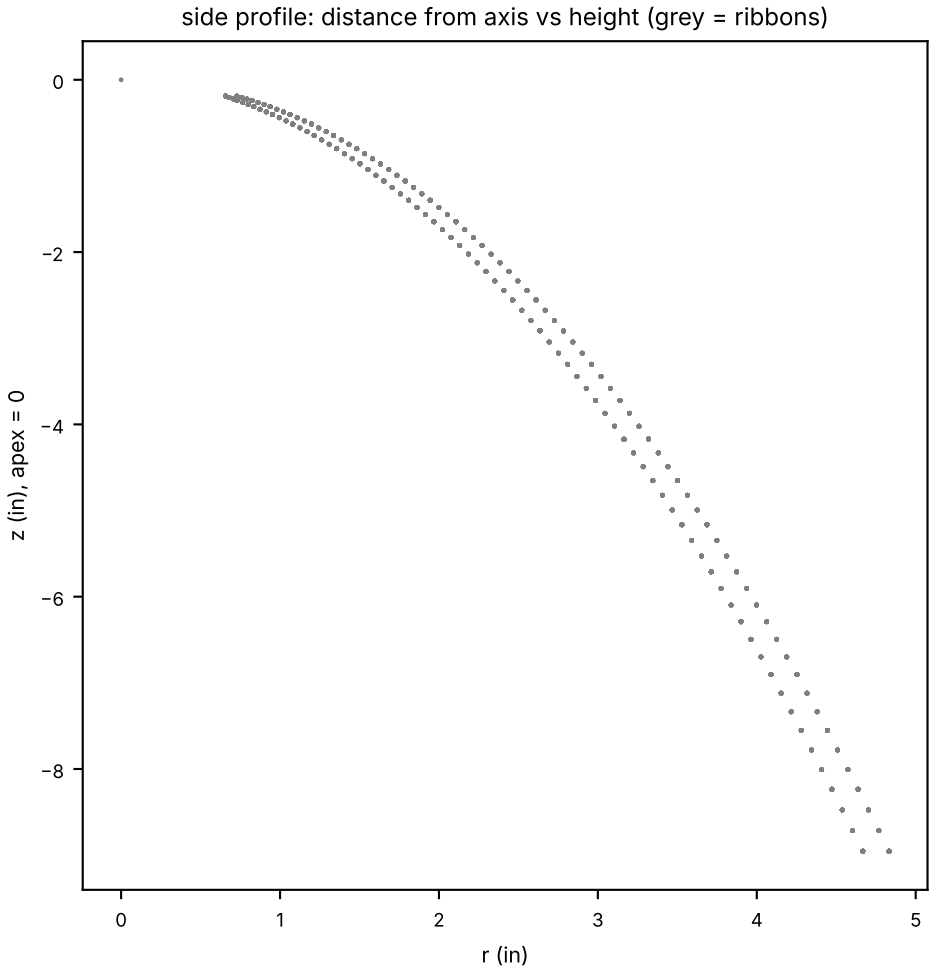




run	build	solver settings	outcome	CPU h	where / result
design_0009	legacy	fineness 4.5; 500 it; default nume	ok	3.24	Cd 0.7364

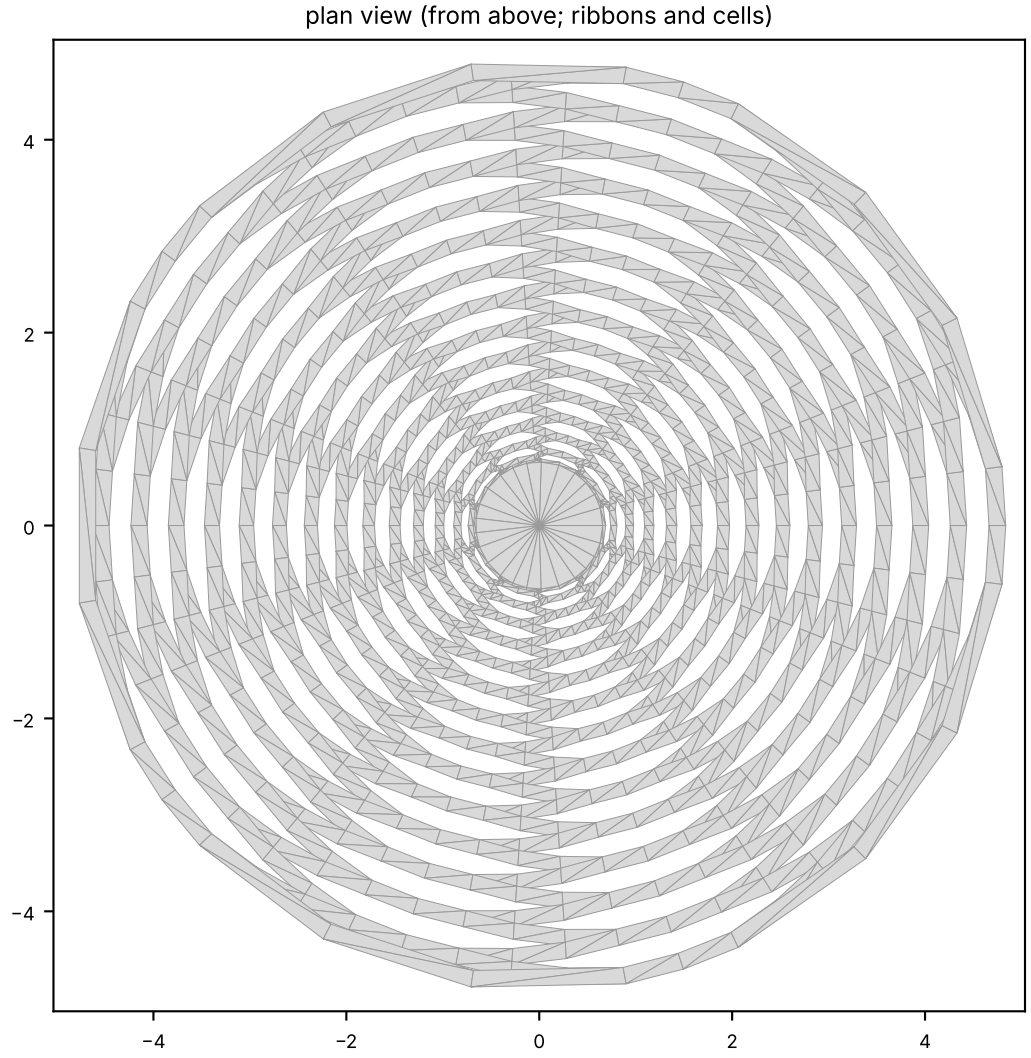
Shape 9: 26 rings, 8 cuts per ring, ring exponent 1.21, inner radius 0.69 in

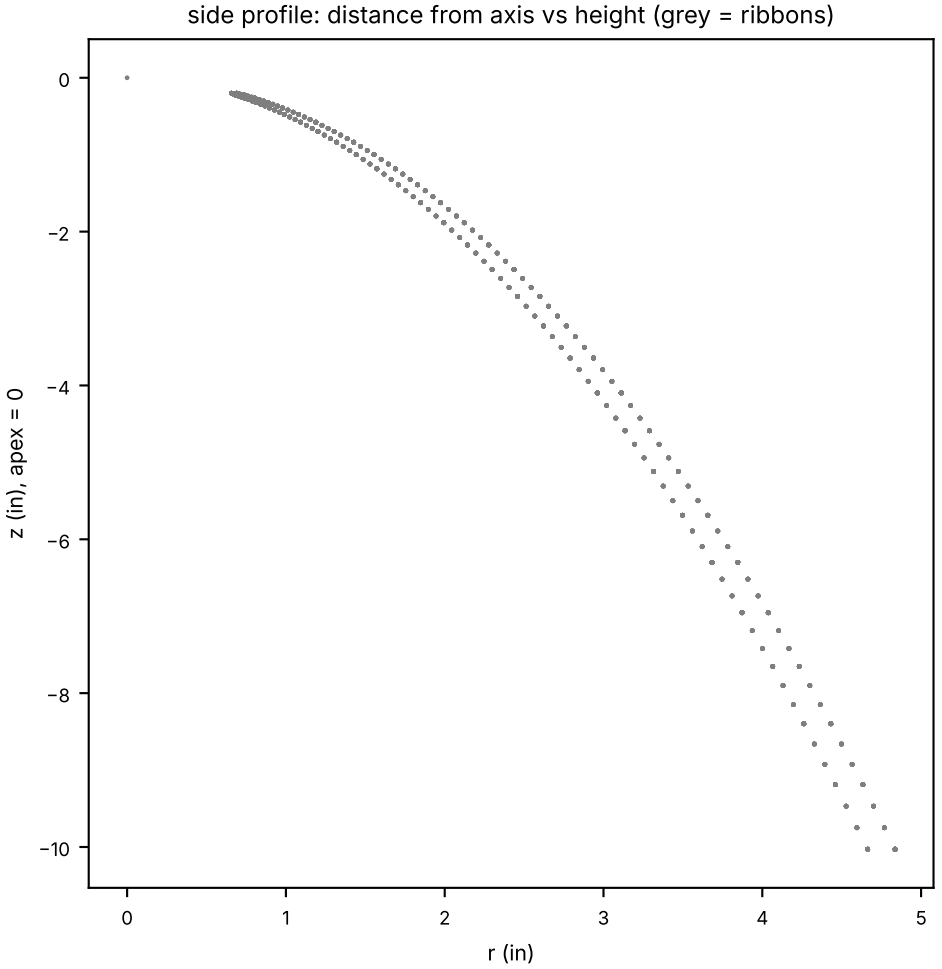




run	build	solver settings	outcome	CPU h	where / result
design_0010	legacy	fineness 4.5; 500 it; default nume	pipeline / API error (not the geometry)	0.00	no position reported
design_0011	legacy	fineness 4.5; 500 it; default nume	ok	3.56	Cd 0.7507

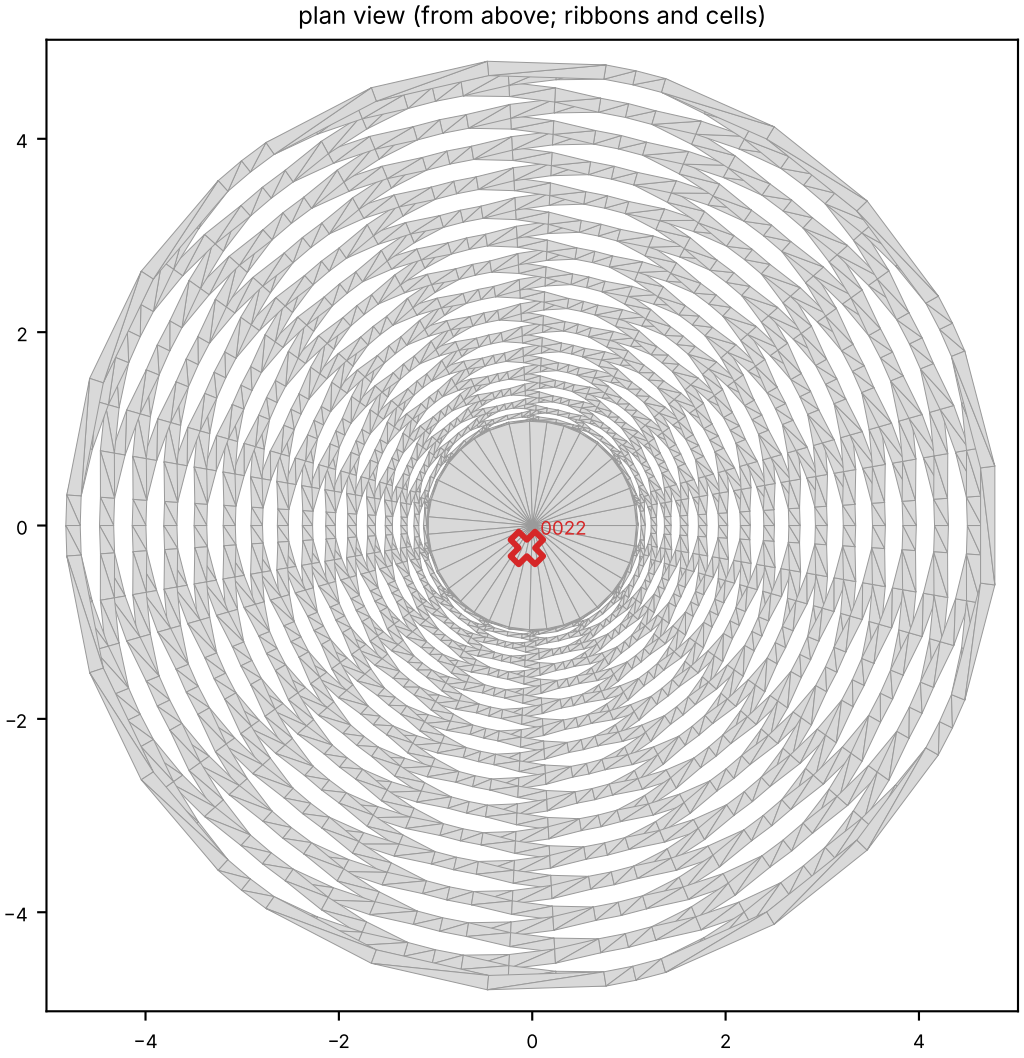
Shape 10: 29 rings, 5 cuts per ring, ring exponent 1.41, inner radius 0.67 in

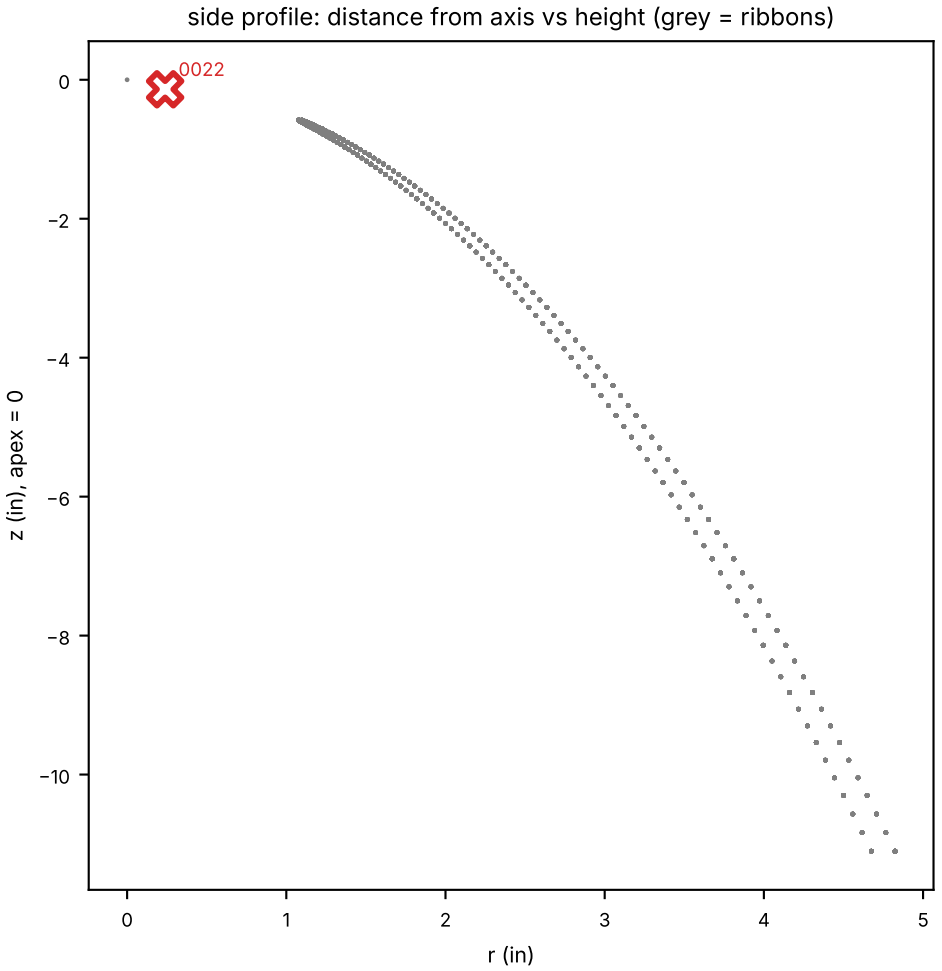




run	build	solver settings	outcome	CPU h	where / result
design_0021	legacy	fineness 4.5; 500 it; default nume	solver diverged (stopped by guard)	1.40	no position reported

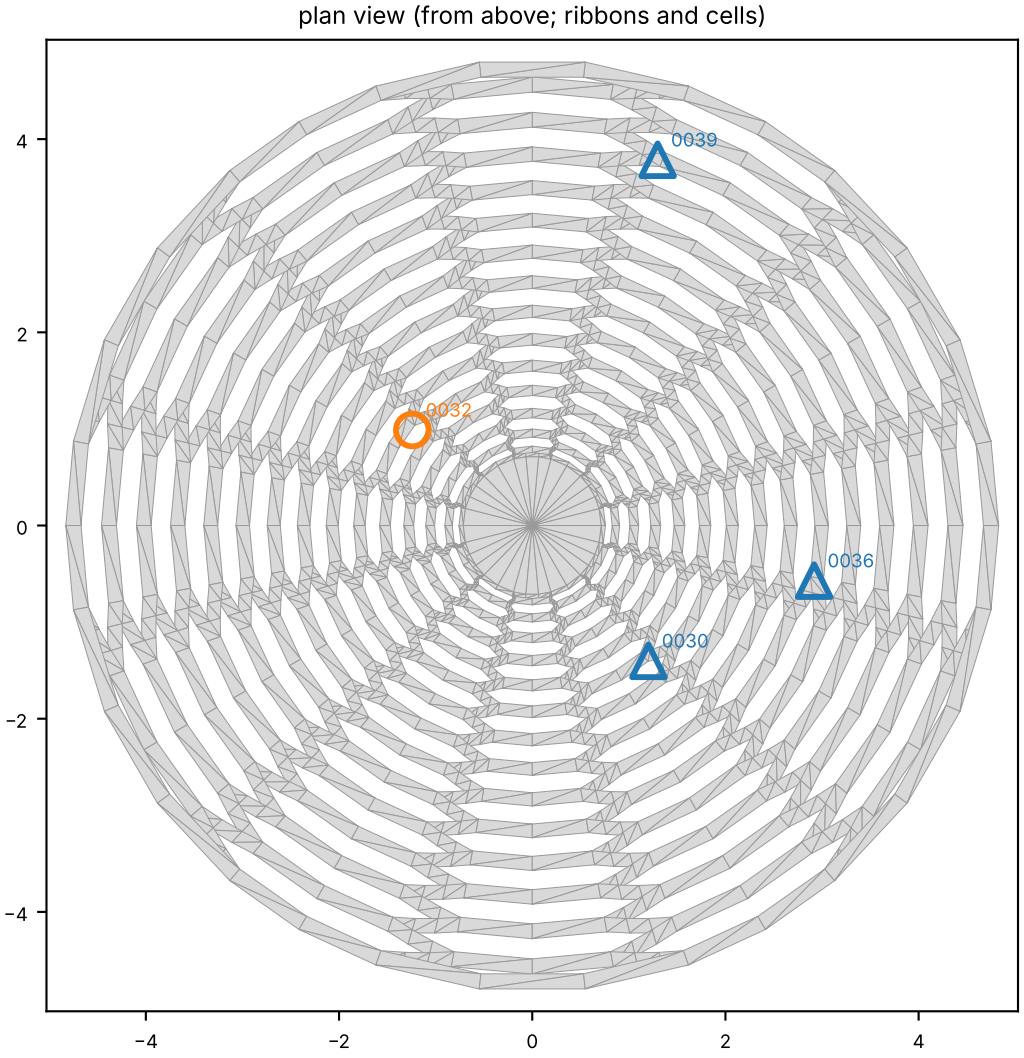
Shape 11: 32 rings, 7 cuts per ring, ring exponent 1.49, inner radius 1.09 in

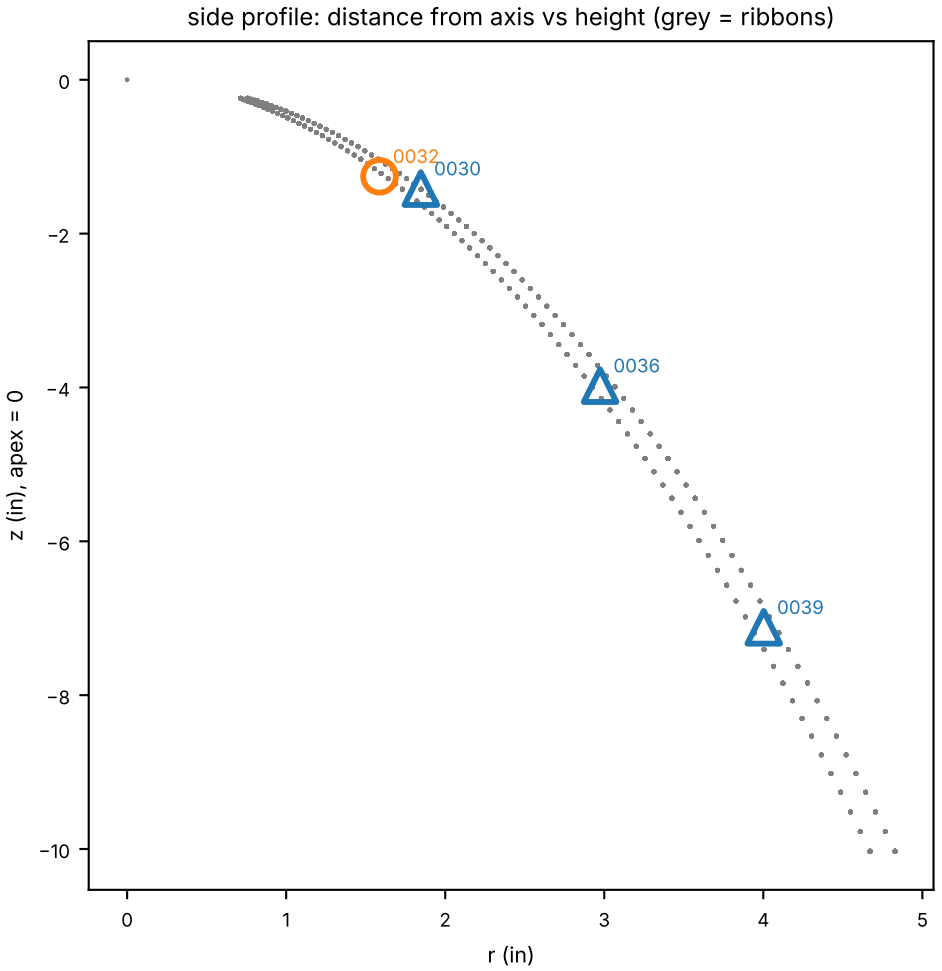




run	build	solver settings	outcome	CPU h	where / result
design_0022	legacy	fineness 4.5; 500 it; default nume	mesher failed	0.00	r 0.24 in, z -0.14 in

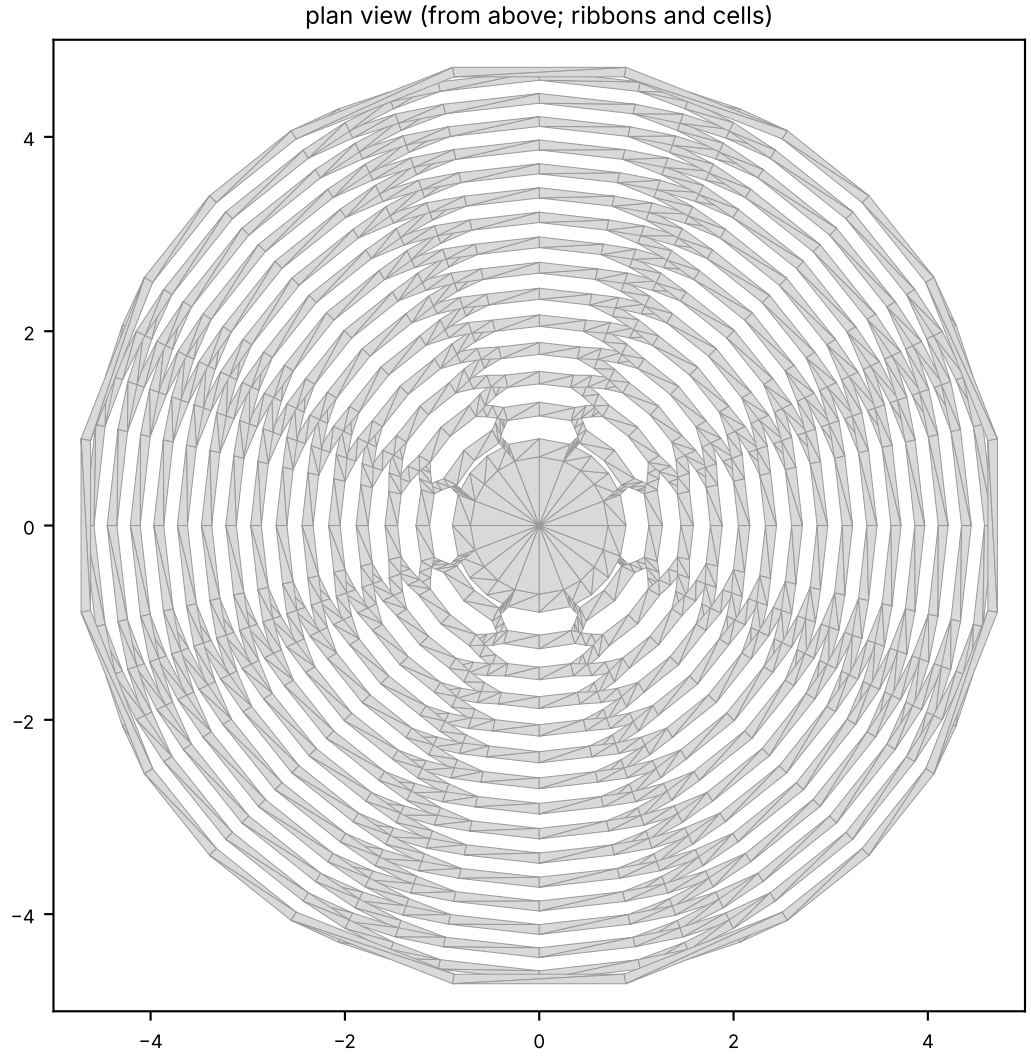
Shape 12: 29 rings, 6 cuts per ring, ring exponent 1.30, inner radius 0.74 in

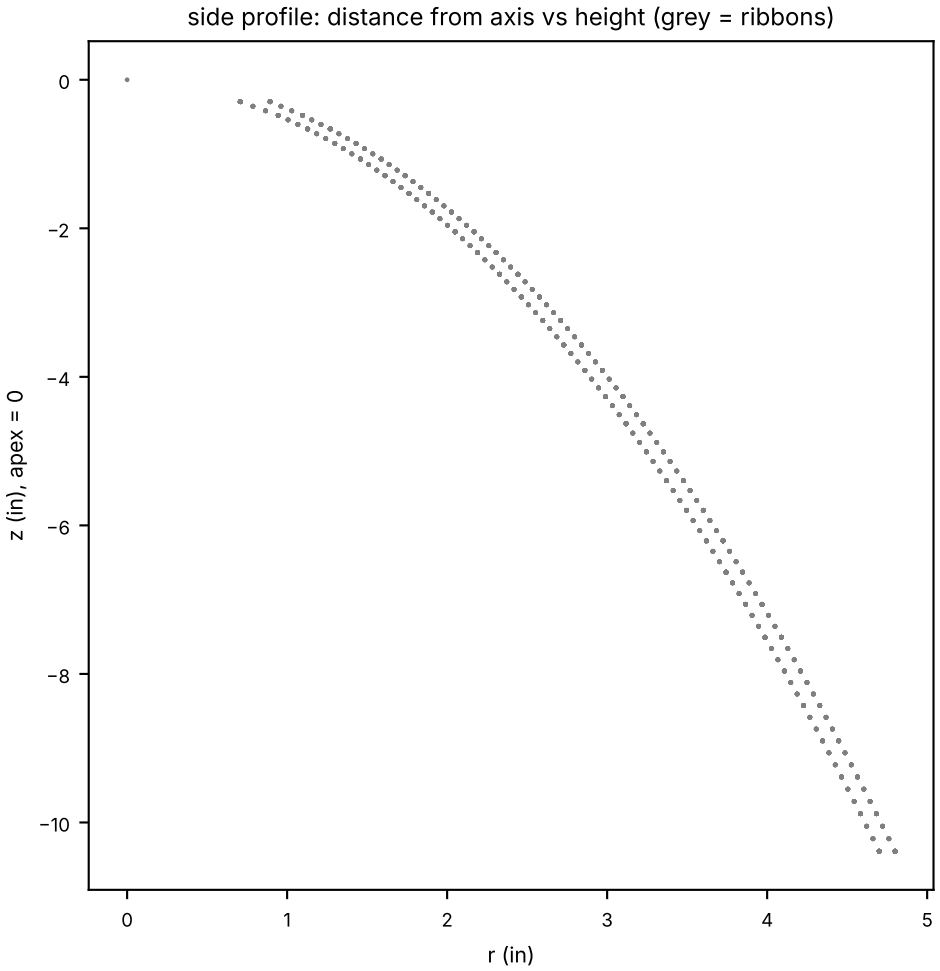




run	build	solver settings	outcome	CPU h	where / result
design_0023	legacy	fineness 4.5; 500 it; default nume	ok	2.93	Cd 0.7863
design_0029	clean	fineness 4.5; 500 it; default nume	mesh cap hit (cancelled)	2.02	no position reported
design_0030	clean	fineness 4.5; 500 it; default nume	solver diverged (stopped by guard)	1.15	r 1.85 in, z -1.41 in
design_0032	clean	fineness 4.5; 500 it; default nume	solver diverged (crashed)	1.11	r 1.59 in, z -1.26 in
design_0034	clean	fineness 6; 500 it; default numeri	solver diverged (stopped by guard)	2.88	no position reported
design_0036	lattice	fineness 4.5; 500 it; default nume	solver diverged (stopped by guard)	1.65	r 2.97 in, z -3.98 in
design_0038	lattice	fineness 4.5; 500 it; potential-fl	solver diverged (stopped by guard)	1.40	no position reported
design_0039	lattice	fineness 4.5; 500 it; default nume	solver diverged (stopped by guard)	1.36	r 4.00 in, z -7.12 in
design_0040	lattice	fineness 4.5; 500 it; default nume	other: SimScaleError: mesh quality gate:	0.94	no position reported
design_0042	lattice	fineness 4.5; 500 it; default nume	other: SimScaleError: simulation run can	5.48	no position reported
design_0043	lattice	fineness 4.5; 500 it; default nume	ok	18.39	Cd 0.6907

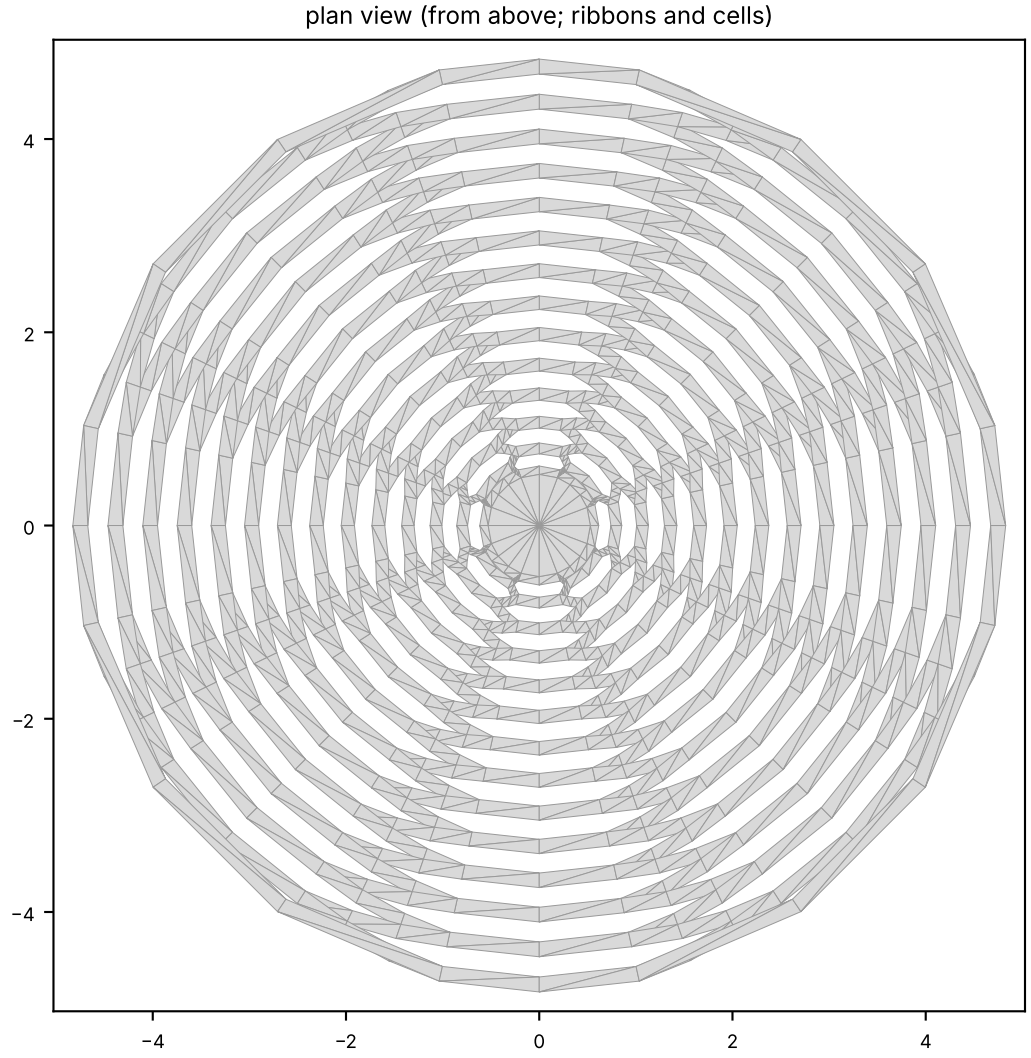
Shape 13: 30 rings, 4 cuts per ring, ring exponent 0.86, inner radius 0.80 in

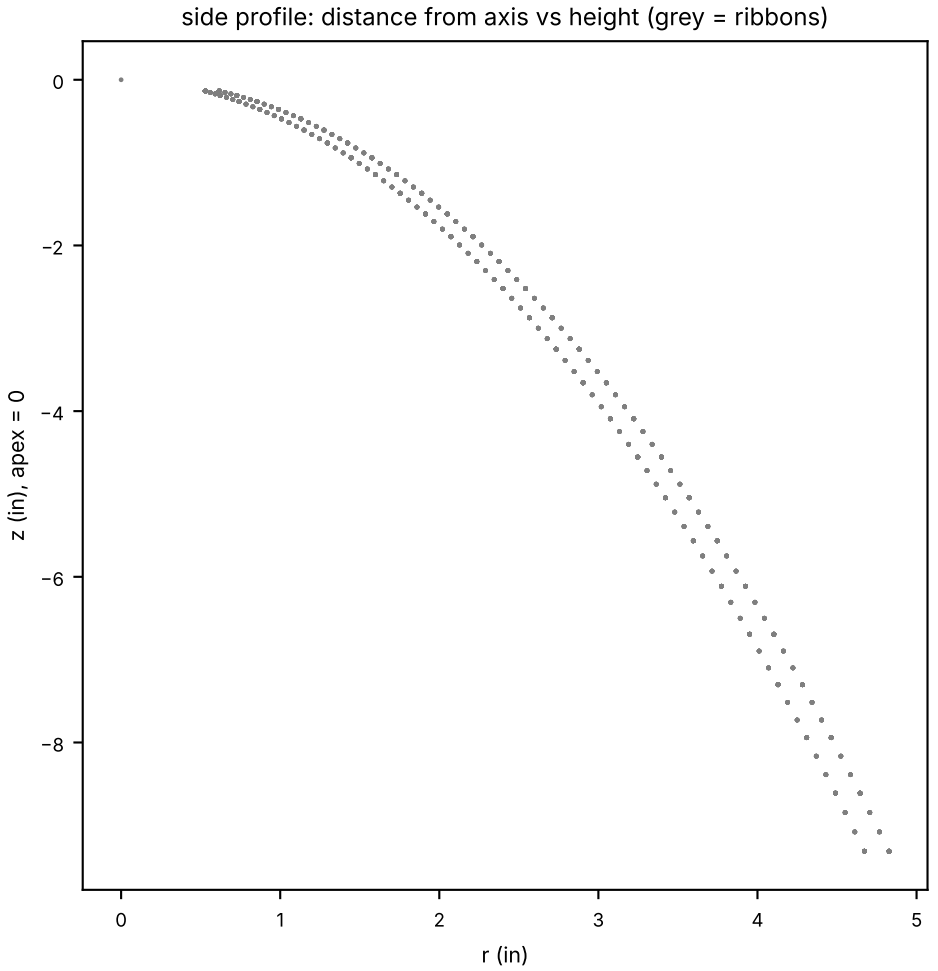




run	build	solver settings	outcome	CPU h	where / result
design_0024	legacy	fineness 4.5; 500 it; default nume	solver diverged (stopped by guard)	2.21	no position reported

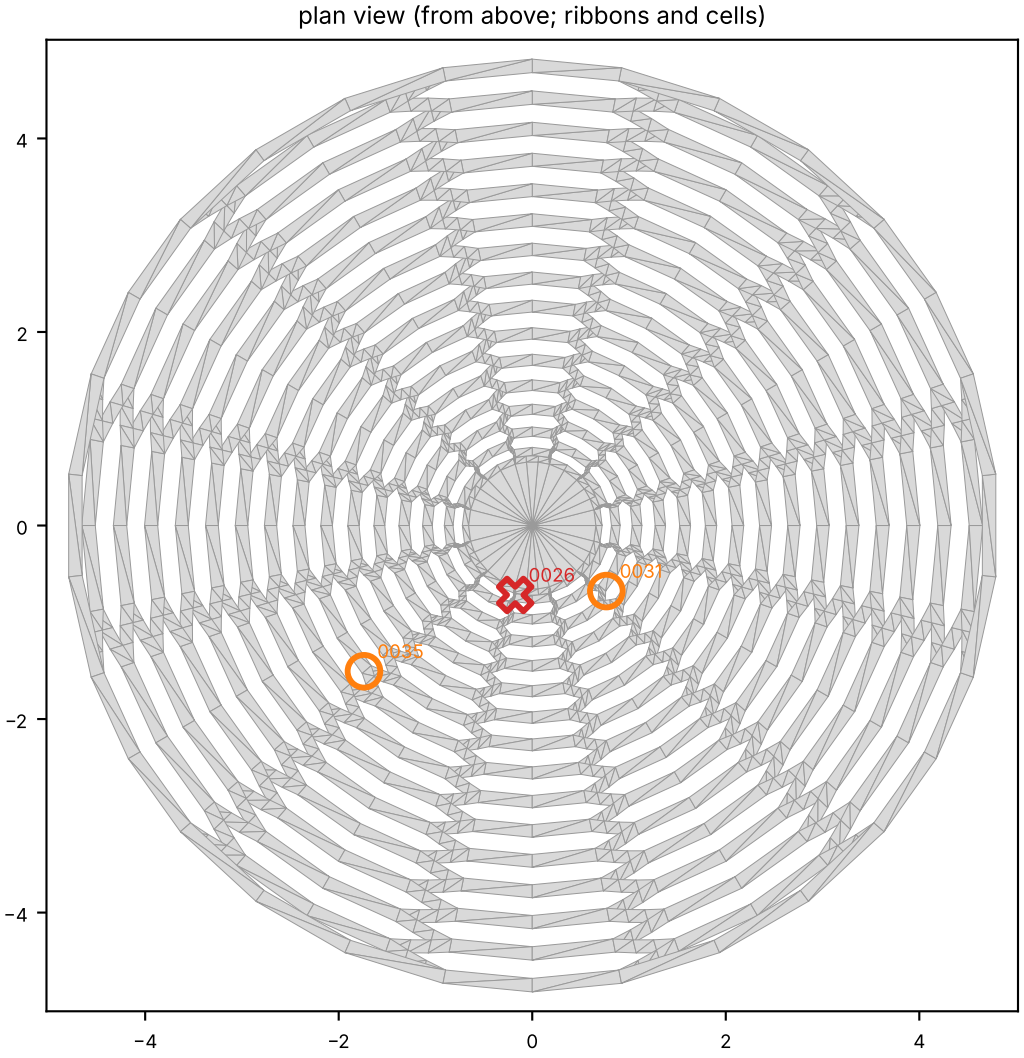
Shape 14: 27 rings, 4 cuts per ring, ring exponent 1.14, inner radius 0.57 in

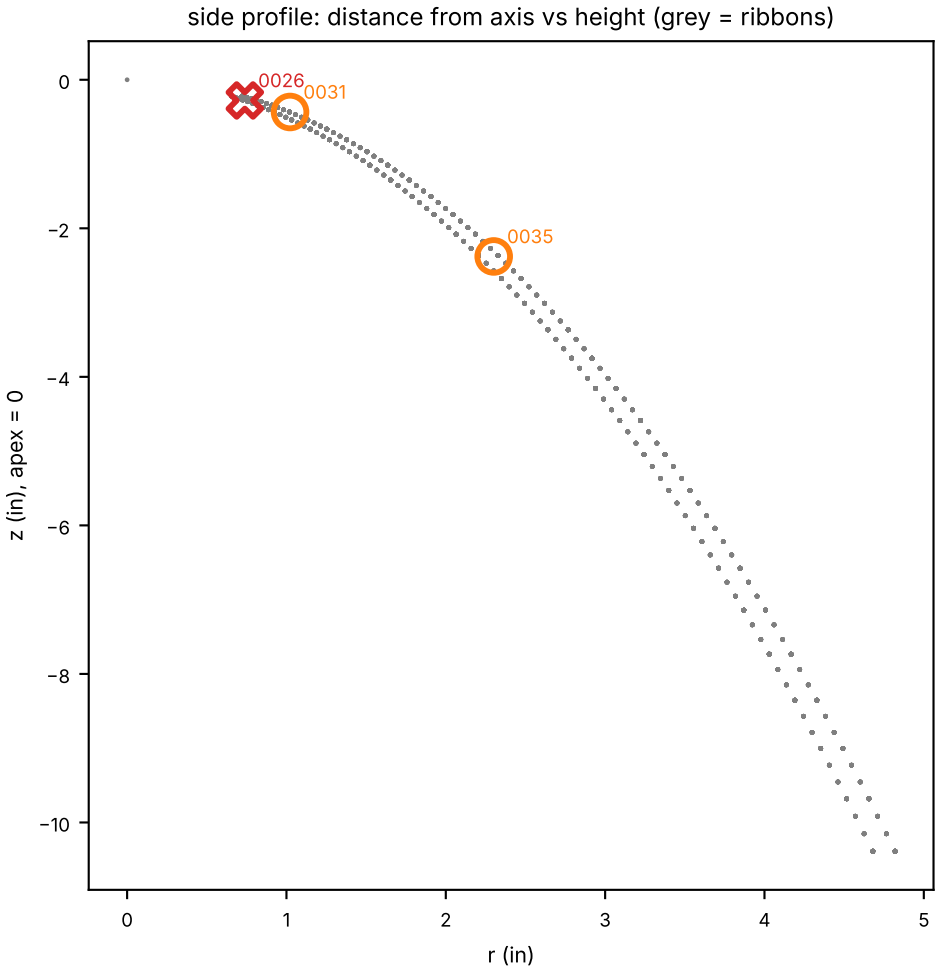




run	build	solver settings	outcome	CPU h	where / result
design_0025	legacy	fineness 4.5; 500 it; default nume	ok	2.71	Cd 0.7457

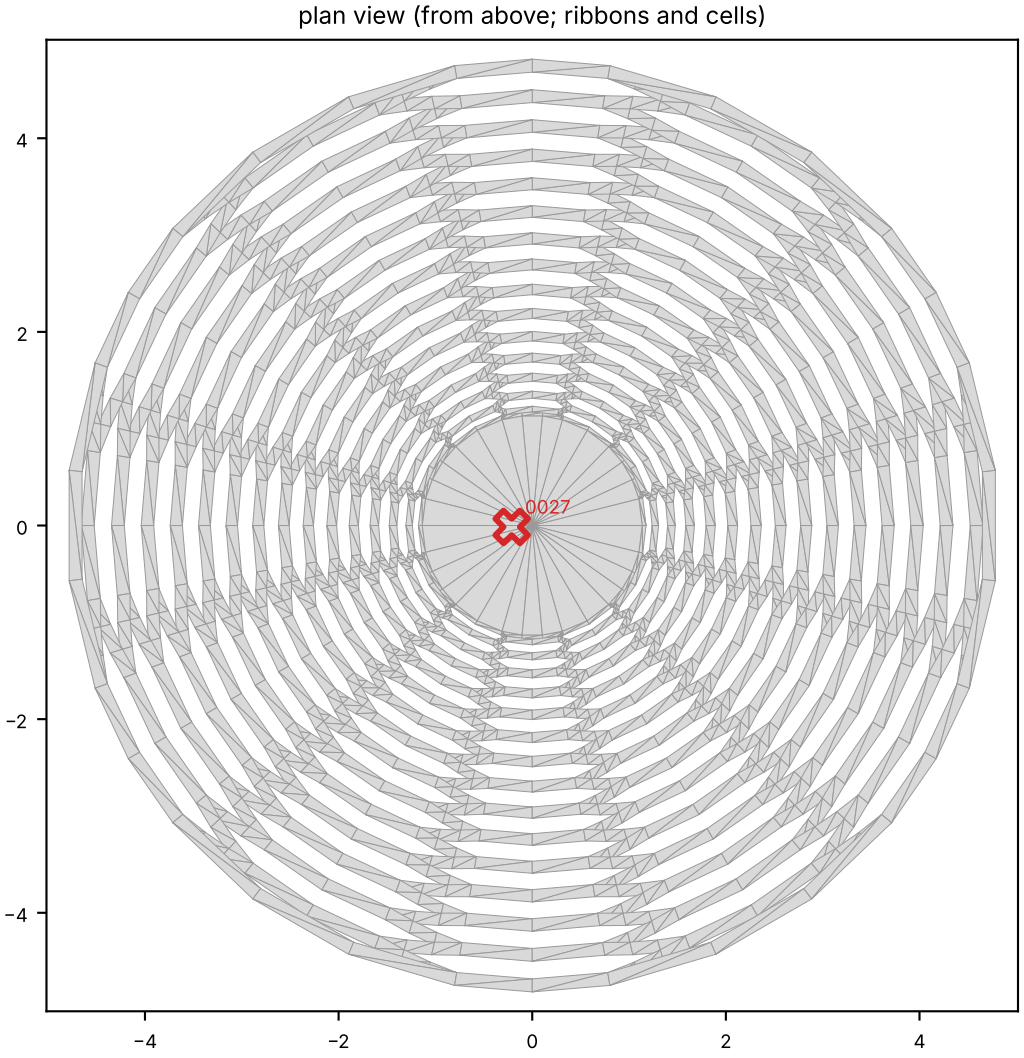
Shape 15: 30 rings, 6 cuts per ring, ring exponent 1.18, inner radius 0.69 in

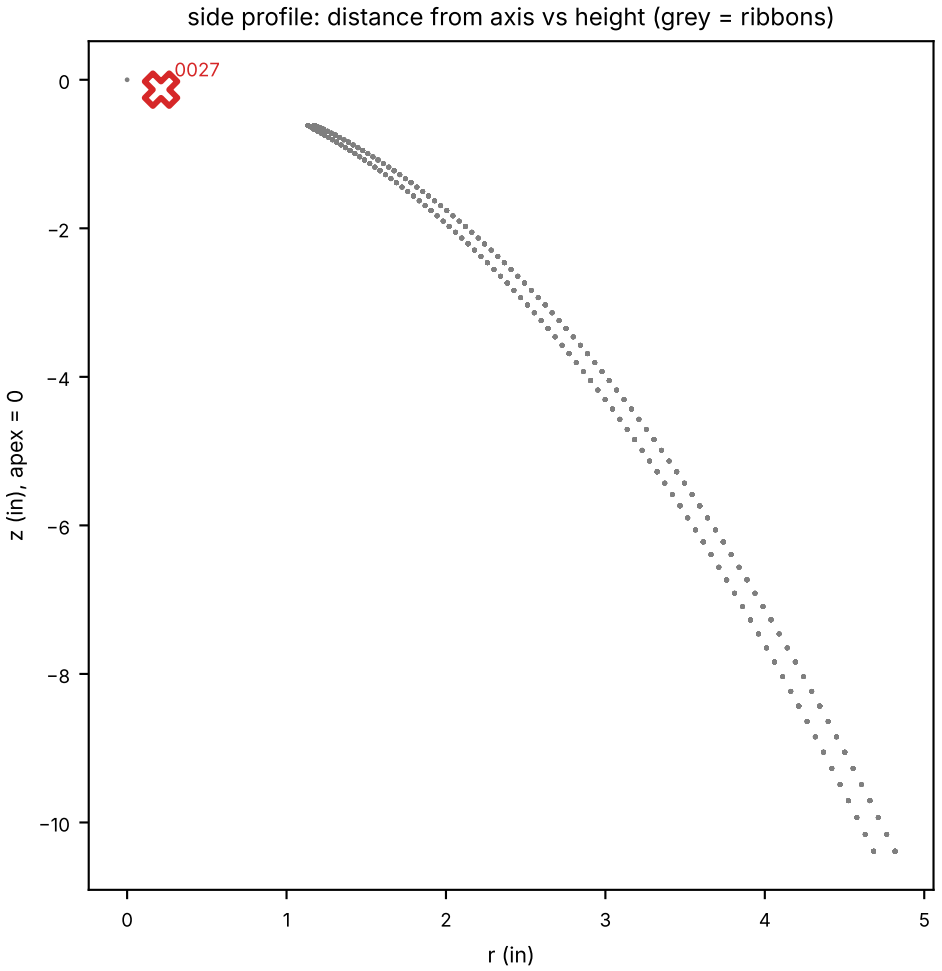




run	build	solver settings	outcome	CPU h	where / result
design_0026	legacy	fineness 4.5; 500 it; default nume	mesher failed	0.00	r 0.74 in, z -0.28 in
design_0031	clean	fineness 4.5; 500 it; default nume	solver diverged (crashed)	0.10	r 1.02 in, z -0.43 in
design_0033	clean	fineness 4.5; 500 it; default nume	solver diverged (stopped by guard)	0.92	no position reported
design_0035	clean	fineness 6; 500 it; default numeri	solver diverged (crashed)	0.12	r 2.30 in, z -2.38 in
design_0037	lattice	fineness 4.5; 500 it; default nume	solver diverged (stopped by guard)	1.43	no position reported
design_0041	lattice	fineness 4.5; 500 it; default nume	other: SimScaleError: mesh quality gate:	0.85	no position reported

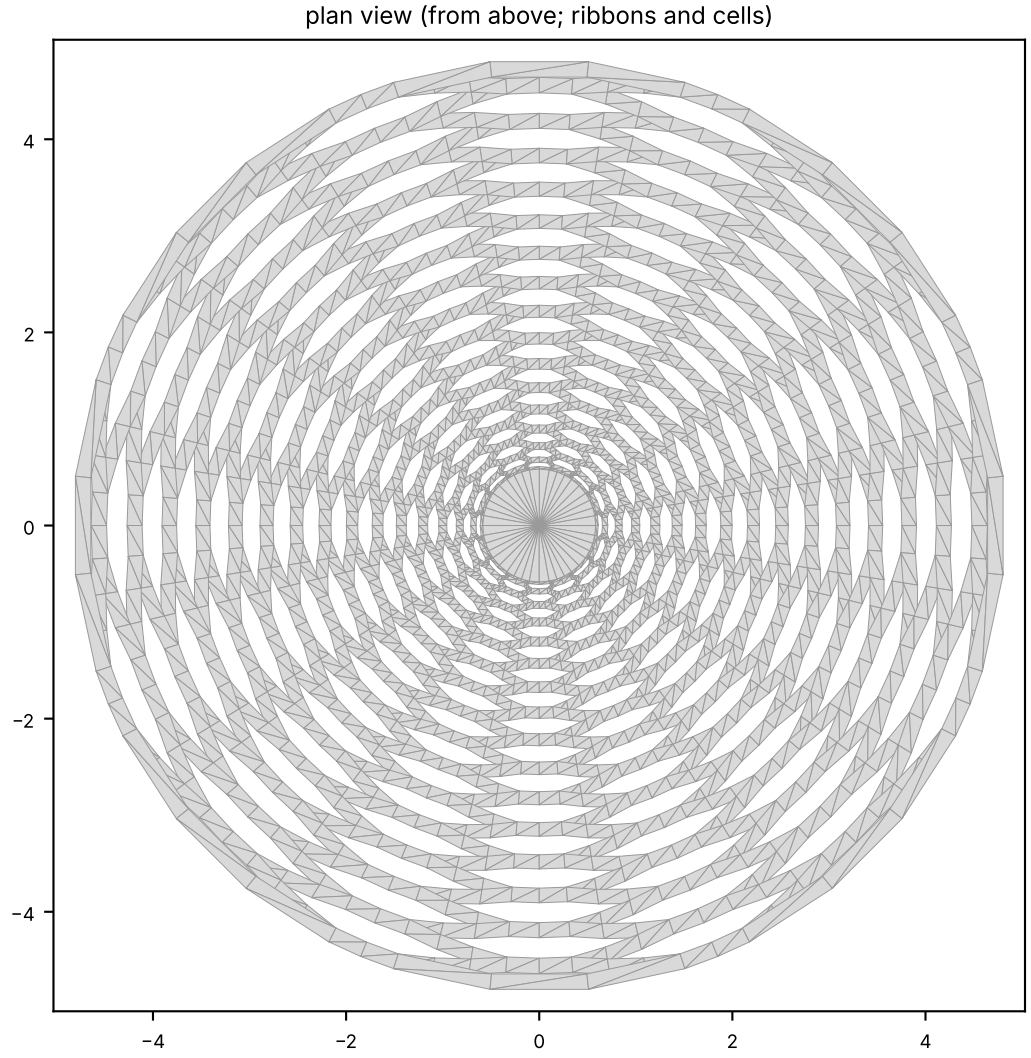
Shape 16: 30 rings, 6 cuts per ring, ring exponent 1.28, inner radius 1.16 in

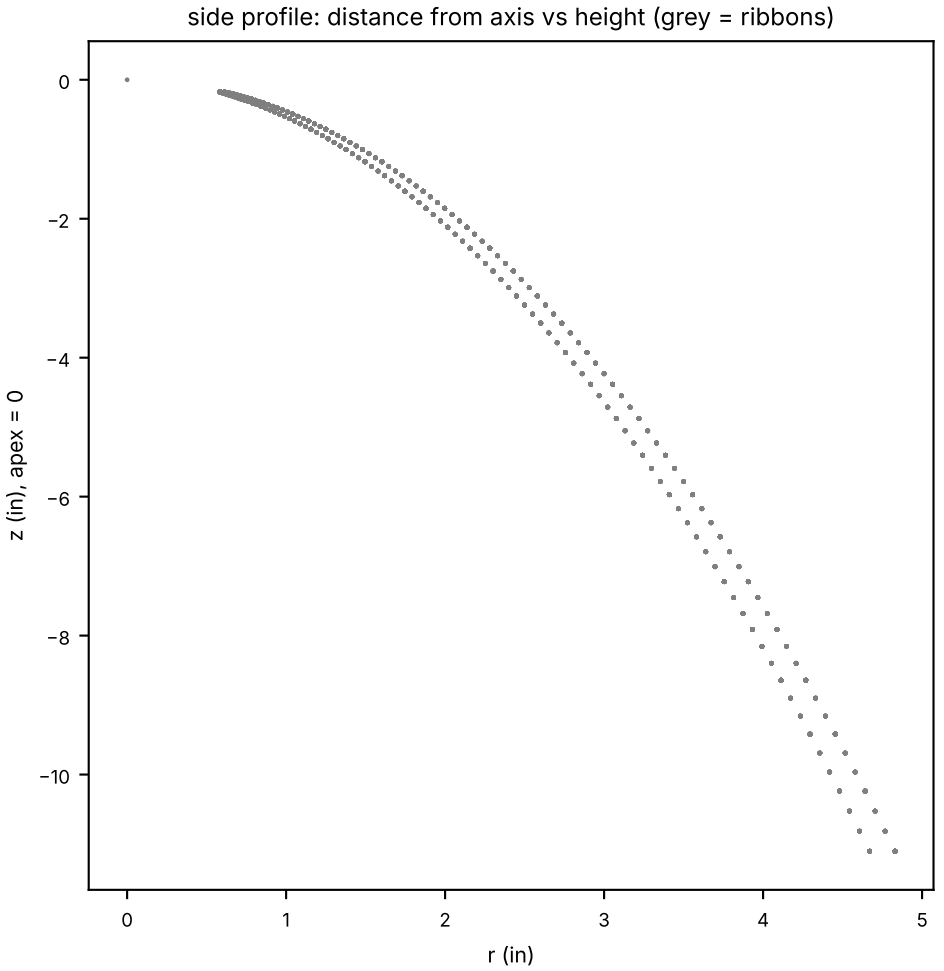




run	build	solver settings	outcome	CPU h	where / result
design_0027	legacy	fineness 4.5; 500 it; default nume	mesher failed	0.00	r 0.21 in, z -0.13 in

Shape 17: 32 rings, 8 cuts per ring, ring exponent 1.42, inner radius 0.60 in





run	build	solver settings	outcome	CPU h	where / result
design_0028	legacy	fineness 4.5; 500 it; default nume	ok	3.89	Cd 0.7316